MATTER & STATES OF MATTER — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 1 · Matter + States + Solids + Mixtures + Separation + Mole Concept + Gas Laws

Ye full notes nahi hai — Chapter 1 ke source mindmap se high-memory/high-confusion facts, values, classifications, separation methods aur gas-law formulas ko fast exam-recall tricks mein compress kiya gaya hai.

1. Matter — Mass + Space

- Matter = anything that has mass and occupies space
- Matter molecules aur atoms se bana hota hai
- Source examples: Water, Air, Iron
- Main particle properties: interstitial space, constant motion, intermolecular attraction

TRICK - "Matter = MASS + SPACE"
Has mass + occupies space.

2. Physical vs Chemical Classification

Classification	Types
Physical	Solid, Liquid, Gas, Plasma, BEC
Chemical	Pure substance → Element/Compound; Mixture → Homogeneous/Heterogeneous

TRICK - "Physical = 5 States; Chemical = Pure vs Mixture"
Pure substance = Element + Compound.
Mixture = Homogeneous + Heterogeneous.

3. Earth Crust + Atmosphere — O-S / N-O

Region	Source Composition
Earth crust	Oxygen 46.8%; Silicon 27.7%; Aluminium 8%; Iron 5%; others
Atmosphere	Nitrogen 78%; Oxygen 21%; CO_2 ≈ 0.03%

TRICK - "Crust = O-S; Air = N-O"
Crust: Oxygen first, Silicon second.
Atmosphere: Nitrogen first, Oxygen second.

EXAM TRAP - "46.8 - 27.7 / 78 - 21"
Crust O-S / Atmosphere N-O number chain.

4. Five States — S-L-G-P-B

State	Shape / Volume / Force	Source Hook
Solid	Definite shape + volume; strong IMF	Negligible compressibility
Liquid	No definite shape; definite volume	More compressible than solid
Gas	No definite shape/volume; weak IMF	High compressibility
Plasma	Ionised gas	4th state; Sun & stars
BEC	Near absolute zero	5th state

TRICK - "S-L-G-P-B = 1-2-3-4-5"
Plasma = 4th; BEC = 5th.

EXAM TRAP -
Plasma 4th, BEC 5th — reverse mat karna.

5. BEC — Bose + Einstein + Cornell

- Source: BEC concept/discovery = Satyendra Nath Bose + Albert Einstein (1924)
- Formed by dilute gas of bosons near absolute zero: 0 K = -273°C
- Source: Eric Cornell (2001) experimentally created BEC using Rubidium-85 and Nobel reference

TRICK - "Bose-Einstein 1924; Cornell 2001; Rb-85"
BEC memory chain: B-E / C / Rb.

6. Phase Changes — M-F-V-C-S-D

Change	Process	Example
Solid → Liquid	Melting / Fusion	Ice → Water
Liquid → Solid	Freezing	Water → Ice
Liquid → Gas	Vaporisation	Water → Steam
Gas → Liquid	Condensation	Steam → Water; dew/fog/mist
Solid → Gas	Sublimation	Camphor, Dry Ice, Naphthalene
Gas → Solid	Deposition	Frost formation

TRICK - "M-F-V-C-S-D"
Melting-Freezing-Vaporisation-Condensation-Sublimation-Deposition.

EXAM TRAP - "Sublime = C-D-N-I"
Camphor - Dry Ice - Naphthalene - Iodine.

7. Evaporation + Boiling Point

Factor	Evaporation Rate
Surface area ↑	Increases
Wind speed ↑	Increases
Temperature ↑	Increases
Humidity ↑	Decreases

TRICK - "S-W-T UP → Evap UP; Humidity UP → Evap DOWN"
Humidity opposite direction mein kaam karti hai.

- Water boiling point at sea level = 100°C
- Altitude ↑ → pressure ↓ → boiling point ↓
- Sea water boiling point > normal water because dissolved salts
- Pressure cooker: pressure ↑ → boiling point ↑

EXAM TRAP - "Altitude LOWERS BP; Pressure Cooker RAISES BP"
Pressure-direction trap.

8. Crystalline vs Amorphous

Feature	Crystalline	Amorphous
Structure	Regular / ordered	Random / disordered
Shape	Definite geometrical	Indefinite
Melting point	Sharp	Range, not sharp
Direction property	Anisotropic	Isotropic
Examples	Quartz, NaCl, Ice	Glass, Plastic, Rubber

TRICK - "Crystal = A-S; Amorphous = I-R"
Crystalline = Anisotropic + Sharp MP.
Amorphous = Isotropic + Range of MP.

EXAM TRAP - "Glass Amorphous; Quartz Crystalline"
Dono source mein SiO_2 forms ke contrast ke roop mein diye gaye hain.

9. Crystalline Solids — I-M-C-M

Type	Example	Bonding / Property
Ionic	NaCl	Electron transfer; high MP; brittle; molten conductor
Molecular	Ice	Weak intermolecular force; low MP; soft; poor conductor
Covalent	Diamond	Covalent network; very hard; very high MP; non-conductor
Metallic	Ag, Fe	Sea of free electrons; conductor; ductile; malleable; lustrous

TRICK - "I-M-C-M = NaCl - Ice - Diamond - Ag/Fe"
Ionic-Molecular-Covalent-Metallic.

EXAM TRAP - "Diamond = Covalent; NaCl = Ionic"
Molecular solids source table mein lowest MP due weak forces.

10. Element vs Compound vs Pure Substance

Concept	Source Recall
Element	One kind of particle; cannot be broken into simpler substances; Fe, Cu, Au, O_2, H_2
Compound	Two or more elements in fixed ratio; definite proportions; H_2O, NaCl, $CaCO_3$
Pure Substance	Fixed composition + characteristic properties; includes elements + compounds

TRICK - "Element = One; Compound = Fixed Combo; Pure = E + C"
Compound components fixed ratio mein.

11. Mass Terms + Mole Concept

Item	Source Value / Formula
1 amu	1.66056 × 10^-24 g
Molecular mass H_2O	18.02 amu
Equivalent weight	Molecular weight / Valency
Mass %	Mass solute ×100 / Total mass solution
No. of moles	Mass(g) / Molecular or Atomic Weight
1 mole gas at STP	22.4 L
Avogadro number	6.022 × 10^23 particles

TRICK - "Mole Triple = Gram Formula Weight + 22.4 L + 6.022×10^{23}"
Mass route, gas-volume route, particle-count route.

EXAM TRAP - "H_2O = 18.02; amu = 1.66056×10^-24 g"
Number pair.

12. Mixtures — Homogeneous vs Heterogeneous

Feature	Homogeneous	Heterogeneous
Composition	Uniform	Non-uniform
Particle size	< 1 nm	> 10 nm in source
Light scattering	No	Tyndall effect
Filtration	Cannot	Can by filtration/settling
Examples	Sugar/salt in water, air	Oil/soil/chalk in water

TRICK - "HOMO = Hidden Particles, No Tyndall; HETERO = Tyndall"
True solution homogeneous → no scattering.

EXAM TRAP - "Tyndall = Milk-Fog-Smoke-Colloid"
Source examples.

13. Separation Methods — E-C-F-S-C-D-C-D

Method	Source Example
Evaporation	Salt from water; dye from ink
Centrifugation	Butter/cream from milk
Separating Funnel	Oil from water
Sublimation	Camphor / naphthalene
Chromatography	Drugs from blood; pigments
Distillation	Homogeneous mixture of two liquids
Crystallisation	Salt from sea water; purifying sugar
Decantation	Sand / insoluble particles from liquid

TRICK - "Butter-CENTRI; Oil-FUNNEL; Pigment-CHROMA; Sea Salt-CRYSTAL"
Four highest-yield separation pairs.

EXAM TRAP - "Salt from Sea Water = Crystallisation"
Source explicitly marks this against distillation/evaporation confusion.

14. Physical vs Chemical Change

Type	Core	Examples
Physical	No new substance; usually reversible	Melting ice, breaking pencil, tearing paper, boiling water
Chemical	New substance; mostly irreversible	Burning sawdust, curd formation, rusting, cooking

TRICK - "Physical = Form Changes; Chemical = New Substance"
P = physical form/state; C = chemical identity change.

EXAM TRAP - "Candle = BOTH"
Wax melts = physical; wax burns to CO_2 + H_2O = chemical.

15. Gas Laws — B-C-G-D

Law	Year	Relation	Constant
Boyle	1622	$P \propto 1/V$; $P_1V_1=P_2V_2$	T, n
Charles	1787	$V \propto T$; V/T = constant	P, n
Gay-Lussac	1809	$P \propto T$; P/T = constant	V, n
Dalton	—	Ptotal = PA + PB + PC + ...	Partial pressures

TRICK - "Boyle P-V; Charles V-T; Gay P-T"
Missing pair tells constant:
Boyle P-V → T constant; Charles V-T → P constant; Gay P-T → V constant.

EXAM TRAP - "1622 - 1787 - 1809"
Boyle - Charles - Gay-Lussac year chain.

TRICK - "Dalton = Total Pressure = Sum of Partial Pressures"
Ptotal = PA + PB + PC + ...

16. Ideal Gas — Master Formula

- Ideal gas source assumption: intermolecular attraction negligible
- State variables: P, V, T, amount n
- Combined ideal gas equation: PV = nRT

TRICK - "Boyle + Charles + Gay → PV = nRT"
Ideal-gas master equation.

17. Final Exam Traps — 12 Red Flags

- Plasma = 4th state; BEC = 5th
- BEC source chain = Bose & Einstein 1924; Cornell 2001; Rubidium-85
- Crystalline = anisotropic + sharp MP; amorphous = isotropic + MP range
- Glass = amorphous; Quartz = crystalline
- Diamond = covalent; NaCl = ionic
- Crust: Oxygen first, Silicon second; atmosphere: Nitrogen first
- Humidity ↑ → evaporation ↓
- Tyndall effect: source ties it to heterogeneous/colloidal mixtures; true solution no Tyndall
- Butter from milk = centrifugation
- Salt from sea water = crystallisation
- Burning candle = both physical + chemical
- Boyle PV; Charles V/T; Gay P/T

EXAM TRAP - "P/BEC - Crystal/Glass - Diamond/NaCl - Crust/Air - Humidity - Tyndall - Butter - SeaSalt - Candle - Gas Laws"
Chapter ke highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

“Matter = mass+space, particles have gaps+motion+attraction → Physical S-L-G-P-B, Plasma4 BEC5; BEC Bose-Einstein1924, Cornell2001 Rb-85 → Crust O46.8-Si27.7; atmosphere N78-O21 → phase M-F-V-C-S-D; sublimation C-D-N-I; evaporation SWT↑ but humidity↑→rate↓; altitude pressure↓ BP↓, cooker pressure↑ BP↑ → Solids: crystalline anisotropic+sharp MP vs amorphous isotropic+range; Glass amorphous, Quartz crystalline; crystalline I-M-C-M = NaCl-Ice-Diamond-Ag/Fe → Chemical class: Element one, Compound fixed ratio, Pure=E+C → Mole: 1amu1.66056×10^-24g, H2O18.02amu, 1mol STP22.4L, Avogadro6.022×10^23 → Mixture: Homo uniform <1nm no Tyndall; Hetero nonuniform source>10nm + Tyndall; Milk/Fog/Smoke colloid → Separation: Butter-centri, Oil-funnel, Pigment-chroma, SeaSalt-crystal; Candle BOTH physical+chemical → Gas laws: Boyle1622 PV/Tconst; Charles1787 V/T/Pconst; Gay1809 P/T/Vconst; Dalton partial pressures; ideal PV=nRT.”

TOP 5: Plasma4/BEC5 | Crystal anisotropic vs Amorphous isotropic | 22.4 L + 6.022×10^23 | Butter-centri/SeaSalt-crystal | Boyle-Charles-Gay = PV / V-T / P-T.

ATOMIC STRUCTURE — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 2 · Laws + Atomic Models + Particles + Isotopes + Quantum Numbers + Electronic Configuration

Ye full notes nahi hai — Chapter 2 ke source mindmap se discoverers, years, atomic models, particle values, iso-terms, quantum numbers aur electronic-configuration rules ko fast exam-recall tricks mein compress kiya gaya hai.

1. Atom — Democritus + Kanada + Dalton

- Atom word Greek origin se: indivisible
- Source: atom concept discoverer = Democritus
- Maharishi Kanada ne atom ko indestructible kaha
- Dalton Atomic Theory = 1808

TRICK - "D-K-D = Democritus - Kanada - Dalton"
Democritus = atom concept; Kanada = indestructible; Dalton = atomic theory 1808.

2. Chemical Laws — L-P-D

Law	Scientist	Year	Memory
Conservation of Mass	Antoine Lavoisier	1789	Reactant mass = Product mass
Definite/Constant Proportion	Joseph Proust	1799	Same compound = same mass ratio
Multiple Proportion	Dalton	1803	Small whole-number ratio

TRICK - "L-P-D = 1789-1799-1803"
Lavoisier → Conservation.
Proust → Definite/Constant.
Dalton → Multiple.

TRICK - "Water H:O = 1:8; CO/CO_2 O-mass = 1:2"
Proust example 1:8; Dalton example 1:2.

EXAM TRAP -
Definite Proportion = Constant Proportion; Multiple Proportion is Dalton, not Proust.

3. Atomic Models — D-T-R-B Timeline

Model	Year	Key Idea
Dalton	1808	Atoms indivisible; same element same mass/properties
Thomson	1898	Watermelon / Plum Pudding; e- embedded in positive sphere
Rutherford	1911	Nuclear model; most atom empty; nucleus has positive charge + most mass
Bohr	1913	Fixed K-L-M-N orbits; fixed energy; quanta on transitions

TRICK - "D-T-R-B = 1808-1898-1911-1913"
Dalton → Thomson → Rutherford → Bohr.

TRICK - "Thomson = Watermelon; Rutherford = Nuclear; Bohr = KLMN"
Three model labels ko ek chain mein lock karo.

EXAM TRAP -
Rutherford gold-foil alpha scattering; Bohr based on Planck quantum theory.

4. Rutherford Model — Empty Space + Nucleus

- Most alpha particles pass through → atom mostly empty space
- Positive charge + most mass nucleus mein concentrated
- Electrons and nucleus electrostatic force se bound
- Alpha particle nucleus hit kare → repelled

TRICK - "Rutherford = EMPTY + NUCLEUS"
Gold-foil alpha scattering ka core conclusion.

5. Stable Particles — NEP

Particle	Discoverer / Year	Charge	Mass	Location
Electron	J.J. Thomson 1897	-1.6022×10^{-19} C	9.1095×10^{-31} kg	Outside nucleus
Proton	Rutherford 1919	$+1.6022\times10^{-19}$ C	1.672×10^{-27} kg	Inside nucleus
Neutron	James Chadwick 1932	0	1.675×10^{-27} kg	Inside nucleus

TRICK - "NEP = Neutron-Chadwick, Electron-Thomson, Proton-Rutherford"
Notice → Neutron → Chadwick.
Every → Electron → Thomson.
Particle → Proton → Rutherford.

EXAM TRAP - "e mass = 1/1836 of proton"
Proton mass ≈ neutron mass.

6. Millikan + Hydrogen + Nucleons

- R.A. Millikan Oil Drop experiment = 1909
- Electron charge = 1.602×10^{-19} C
- Hydrogen nucleus = only proton, no neutron — source special note
- Protons + Neutrons inside nucleus = Nucleons
- De-Broglie: particles have dual nature = wave + particle

EXAM TRAP - "Millikan = Charge; Thomson = Electron"
Discovery of electron vs measurement of charge ko mix mat karo.

7. Unstable Particles — P-M-N-Ap-An

Particle	Discoverer	Year	Source Mass Hook
Positron	C.D. Anderson	1932	= electron mass
Meson	Yukawa	1935	200× electron mass
Neutrino	Pauli	1927	Indefinite
Anti-Proton	Segre	1955	= proton mass
Anti-Neutron	Cork	1956	= neutron mass

TRICK - "Pauli 1927 Neutrino; Anderson 1932 Positron; Yukawa 1935 Meson"
Early unstable-particle year chain.

8. Atomic Number vs Mass Number — Z vs A

Symbol	Meaning	Formula
Z	Atomic Number	Z = P = e- in neutral atom
A	Mass Number	A = P + N
N	Neutrons	N = A - Z

TRICK - "Z = Proton; A = All Nucleons"
Neutral atom mein Z = proton = electron.
A = proton + neutron.

TRICK - "Carbon: Z6, A12 → P6, e6, N6"
Fast worked example.

EXAM TRAP -
Mass number A = P+N, NOT electrons.

9. Iso-Family — Z-A-N-e

Type	Same	Different	Example
Isotopes	Z	A	1H, 2H, 3H
Isobars	A	Z	Ar-40, K-40, Ca-40
Isotones	N	Z and A	H-3 and He-4: source says both 2 neutrons
Isoelectronic	Electrons	Z	Ne, Na+, Mg2+, Al3+ = 10 e-

TRICK - "IsoTOPE = same Z; IsoBAR = same A; IsoTONE = same N"
T-Z / B-A / Tone-N.

TRICK - "IsoELECTRONIC = Same e-"
Ne, Na+, Mg2+, Al3+ → 10 electrons.

EXAM TRAP -
Isotope same Z; Isobar same A; Isotone same N — classic swap trap.

10. Special Isotope Facts — Source Recall

- Hydrogen isotopes: Protium 1H, Deuterium 2H, Tritium 3H
- Source states Tritium (3H) = lightest isotope and radioactive
- Source states heaviest stable isotope = Lead-208
- Source states element with most isotopes = Polonium (Po)

EXAM TRAP - "H = P-D-T; T Radioactive; Pb-208 Heavy Stable; Po Most"
Ye source-note special-fact chain hai.

11. Four Quantum Numbers — n-l-m-s

Quantum No.	Meaning	Values
n Principal	Energy level + distance	1,2,3,4... = K,L,M,N
l Azimuthal	Orbital shape	0 to n-1; s0 p1 d2 f3
m Magnetic	Orientation	-l to +l
s Spin	Electron spin direction	+1/2 or -1/2

TRICK - "n-l-m-s = Level-Shape-Orientation-Spin"
n = level; l = shape; m = orientation; s = spin.

EXAM TRAP - "s-p-d-f = 0-1-2-3"
Azimuthal quantum number values.

12. Shell Capacity — $2n^2$

Shell	n	Max e-
K	1	2
L	2	8
M	3	18
N	4	32

TRICK - “K-L-M-N = 2-8-18-32”
Formula = $2n^2$.

13. Aufbau — Lower Energy First

- Aufbau = German “construction/building up”
- Electrons fill lower-energy orbitals first
- Source order: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d

EXAM TRAP - “4s BEFORE 3d”
Aufbau ka most common exam trap.

14. Cr & Cu — Aufbau Exceptions

Element	Expected	Actual	Why
Cr 24	3d4 4s2	3d5 4s1	Half-filled d more stable
Cu 29	3d9 4s2	3d10 4s1	Fully-filled d more stable

TRICK - “Cr = d5s1; Cu = d10s1”
Half-filled / fully-filled stability.

15. Pauli vs Hund

Rule	Core
Pauli Exclusion	No two electrons have identical 4 quantum numbers; max 2 e-/orbital; opposite spins
Hund Maximum Multiplicity	Equal-energy orbitals fill singly first, then pairing

TRICK - “Pauli = Pair Opposite; Hund = Single First”
Pauli limits an orbital to 2 opposite-spin electrons.
Hund says spread first, pair later.

EXAM TRAP - “p3 = ↑ ↑ ↑ ; p6 = ↑↓ ↑↓ ↑↓”
Half-filled and fully-filled patterns.

16. Discoverer + Year Flash

Fact	Person	Year
Conservation Mass	Lavoisier	1789
Definite Proportion	Proust	1799
Multiple Proportion	Dalton	1803
Atomic Theory	Dalton	1808
Electron	J.J. Thomson	1897
Thomson Model	J.J. Thomson	1898
Millikan Oil Drop	R.A. Millikan	1909
Rutherford Model	Rutherford	1911
Bohr Model	Niels Bohr	1913
Proton	Rutherford	1919
Neutron	James Chadwick	1932

TRICK - "1789-99-1803 / 1897-98 / 1909-11-13-19-32"
Laws → Thomson → Millikan/Rutherford/Bohr/Proton/Neutron.

17. Final Exam Traps — 12 Red Flags

- Thomson = Watermelon/Plum Pudding, not nuclear model
- Rutherford = nuclear model + source proton discovery 1919
- Bohr = fixed K-L-M-N orbits; based on Planck quantum theory
- Electron = Thomson 1897; electron charge = Millikan 1909
- Electron mass = 1/1836 proton mass
- Hydrogen source special note: no neutron
- Atomic number Z = protons; mass number A = P+N
- Isotope same Z; Isobar same A; Isotone same N
- Source special: most isotopes = Polonium; heaviest stable = Pb-208
- Quantum numbers n-l-m-s = level-shape-orientation-spin
- Aufbau: 4s before 3d; Cr/Cu exceptions
- Pauli = max 2 opposite spins; Hund = single first

EXAM TRAP - "Thomson-Rutherford-Bohr-Millikan-1836-H-Z/A-Iso-Po/Pb-nlms-4s/CrCu-Pauli/Hund"
Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain
"Atom: Democritus concept, Kanada indestructible, Dalton1808 → Laws L-P-D = Lavoisier1789 Conservation, Proust1799 Definite/Constant, Dalton1803 Multiple → Models D-T-R-B = 1808/1898/1911/1913: Thomson Watermelon, Rutherford Nuclear via alpha-gold foil, Bohr KLMN + quanta → Particles NEP: Neutron-Chadwick1932, Electron-Thomson1897, Proton-Rutherford1919; e charge1.602×10^-19C, e mass1/1836 proton; Millikan1909 charge; H nucleus no neutron, P+N=Nucleons → Z=P=e(neutral), A=P+N, N=A-Z → Iso: isotope sameZ, isobar sameA, isotone sameN, isoelectronic same e; H=P-D-T, source T radioactive, Pb208 heavy stable, Po most isotopes → Quantum n-l-m-s = level-shape-orientation-spin; s-p-d-f=0-1-2-3; shell $2n^2$ = K2 L8 M18 N32 → Aufbau lower-energy first, 4s before3d; Cr3d5 4s1, Cu3d10 4s1 exceptions → Pauli max2 opposite-spin per orbital; Hund single-first then pair."
TOP 5: L-P-D laws | Thomson-Watermelon / Rutherford-Nuclear / Bohr-KLMN | NEP particle discoverers | Iso Z-A-N | 4s before 3d + Cr/Cu exceptions.

PERIODIC CLASSIFICATION OF ELEMENTS — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 3 · Dobereiner + Newland + Mendeleev + Moseley + Groups/Periods + Blocks + Periodic Trends

Ye full notes nahi hai — Chapter 3 ke source mindmap se scientists/years, old vs modern periodic table, groups-periods, s-p-d-f blocks aur periodic-property trends ko fast exam-recall tricks mein compress kiya gaya hai.

1. History Timeline — D-N-M-M

Stage	Scientist	Year	Basis / Hook
Triads	Wolfgang Dobereiner	1829	Groups of 3; atomic mass
Octaves	John Newland	1864	Every 8th repeats; atomic mass
Periodic Table	Dmitri Mendeleev	1869	Atomic mass
Modern Table	Henry Moseley	1913	Atomic number

TRICK - "D-N-M-M = 1829-1864-1869-1913"
Dobereiner → Newland → Mendeleev → Moseley.

EXAM TRAP - "Mass-Mass-Mass → Number"
Dobereiner, Newland, Mendeleev = atomic mass; Modern/Moseley = atomic number.

2. Dobereiner Triads — Middle = Average

- Three elements with similar properties arranged by increasing atomic weight
- Middle element mass ≈ average of outer two
- Formula: Middle mass = (1st + 3rd)/2
- Failed because all elements could not be fitted into triads

Triad	Outer Masses	Middle	Check
Li-Na-K	7 & 39	Na 23	(7+39)/2 = 23
Ca-Sr-Ba	40 & 137	Sr 87.5	Average ≈ 88.5
Cl-Br-I	35.5 & 127	Br 80	Average ≈ 81.25

TRICK - "Triad = THREE; Middle = MEAN"
Li-Na-K sabse asked calculation.

EXAM TRAP -
Dobereiner = triads of 3, not octaves of 8.

3. Newland Octaves — Every 8th

- John Newland, England, 1864
- Elements increasing atomic mass mein arranged
- Every 8th element properties repeat
- Musical-note analogy: Sa-Re-Ga-Ma-Pa-Da-Ni
- Applied only up to Calcium; 56 elements known
- Source says noble-gas discovery ke baad scheme failed

TRICK - "Newland = Note = 8"
Every 8th repeats; up to Calcium.

EXAM TRAP -
Every 8th, not 7th or 9th; basis atomic mass, not atomic number.

4. Mendeleev — 1869 / 63 / 7 / 9

Fact	Source Recall
Year	1869
Known elements	63
Periods	7
Groups	9
Basis	Atomic mass
Noble gas group	Absent; zero group absent
Hydrogen / Isotopes	Position not fixed
Title	Father of Periodic Table

TRICK - “Mendeleev = 63-7-9”
63 elements, 7 periods, 9 groups.

TRICK - “Mass + Missing Spaces”
Atomic mass basis; blank spaces left for undiscovered elements.

5. Eka Elements — B-A-S → S-G-G

Predicted Name	Modern Element	Symbol
Eka-Boron	Scandium	Sc (21)
Eka-Aluminium	Gallium	Ga (31)
Eka-Silicon	Germanium	Ge (32)

TRICK - “B-A-S → S-G-G”
Eka-Boron→Scandium; Eka-Aluminium→Gallium; Eka-Silicon→Germanium.

TRICK - “Eka = Sanskrit ‘one’”
One step below in group — source note.

EXAM TRAP -
Eka-Aluminium = Gallium; Eka-Silicon = Germanium.

6. Mendeleev vs Modern

Feature	Mendeleev 1869	Modern / Moseley 1913
Basis	Atomic Mass	Atomic Number
Periods	7	7
Groups	9	18
Elements	63	118
Noble gases	Absent / zero group absent	Group 18
H & Isotopes	Position not fixed	Fixed positions

TRICK - “M = Mass + 9; Modern = Number + 18”
Mendeleev vs Moseley ka core contrast.

EXAM TRAP -
Modern table is based on atomic NUMBER, not atomic mass.

7. Modern Table — 118 / 18 / 7

- Henry Moseley, 1913
- Modern periodic law: properties are periodic function of atomic number
- 118 elements, 18 groups, 7 periods
- Source: 11 gaseous elements at normal atmospheric conditions

TRICK - "Modern = 118-18-7"
Elements-Groups-Periods.

8. Important Groups — 1,2,15,16,17,18

Group	Name	Block
1	Alkali Metals	s
2	Alkaline Earth Metals	s
3-12	Transition Elements	d
15	Pnictogens	p
16	Chalcogens	p
17	Halogens	p
18	Noble / Inert Gases	p

TRICK - "1 Alkali, 2 Alkaline; 15 Pnic, 16 Chalc, 17 Halo, 18 Noble"
Group-name ladder.

EXAM TRAP -
Noble gases Group 18 = p-block, separate block nahi.

9. Lanthanoids vs Actinoids

Series	Atomic No.	Period	Block	Source Hook
Lanthanoids	58-71	6	f	Inner transition; rare earth
Actinoids	90-103	7	f	Inner transition; most radioactive/synthetic

TRICK - "Lantha 58-71 / Acti 90-103"
Period 6 / Period 7; both f-block.

EXAM TRAP -
f-block = Lanthanoids + Actinoids.

10. Period Lengths — 2-8-8-18-18-32-32

Period	Elements	Type / Orbitals
1	2	Very short; 1s
2	8	Short; 2s,2p
3	8	Short; 3s,3p
4	18	Long; 4s,3d,4p
5	18	Long; 5s,4d,5p
6	32	Very long; 6s,4f,5d,6p
7	32	Incomplete; 7s,5f,6d,7p

TRICK - "2-8-8-18-18-32-32"
Periods 1→7 element-count chain.

EXAM TRAP - "Ti Z22 = Group 4, Period 4"
Source specific placement fact.

11. Blocks — s-p-d-f

Block	Groups / Series	Examples
s	Groups 1-2	Li, Na, K, Mg, Ca
p	Groups 13-18	C, N, O, F, Cl, Ne
d	Groups 3-12	Fe, Cu, Zn, Cr, Ni
f	Lanthanoids + Actinoids	Ce, Nd, U, Th

TRICK - “s 1-2; p 13-18; d 3-12; f inner”
Block-group master line.

EXAM TRAP - “Hydrogen = s-block Group 1 but NON-METAL”
Source special trap.

12. Periodic Trends — Down vs Across

Property	Down Group	Across Period →	Extreme
Electronegativity	Decreases	Increases	Highest F
Electron Affinity	Decreases	Increases	Highest Cl
Atomic Size	Increases	Decreases	Largest Cs; smallest F
Ionisation Energy	Decreases	Increases	Noble gases very high
Metallic Character	Increases	Decreases	Lose e- more easily down group

TRICK - “Down: Size+Metal UP; EN-EA-IE DOWN”
Group down mnemonic.

TRICK - “Across: Size+Metal DOWN; EN-EA-IE UP”
Period across = mostly opposite.

EXAM TRAP - “F = Electronegativity; Cl = Electron Affinity”
Most common confusion pair.

13. Valency + Noble Gas

- Source: valency = number of electrons for bonding
- Group elements have same valency pattern
- Source says period across valency increases
- Noble gases Group 18: zero valency + very high ionisation energy
- Source says Groups 2, 15, 18 have zero or positive electron affinity

TRICK - “Noble = 0 Valency + High IE”
Group 18 direct recall.

14. Special Important Facts

Question	Source Answer
Smallest Element	H — Hydrogen
Largest Element	Cs — Caesium
Lightest Metal	Li — Lithium
Highest Melting Point	W — Tungsten
Most Electronegative	F — Fluorine
Highest Electron Affinity	Cl — Chlorine
Father of Periodic Table	Dmitri Mendeleev

TRICK - “H-Cs-Li-W-F-Cl”
Smallest-Largest-Lightest metal-Highest MP-Most EN-Highest EA.

EXAM TRAP -
Lightest metal = Li, not H; H is non-metal.

15. Comparison Flash — Person / Year / Basis

Person	Year	Basis	Group Hook
Dobereiner	1829	Atomic Mass / triads	3-element sets
Newland	1864	Atomic Mass / octaves	Every 8th
Mendeleev	1869	Atomic Mass	9 groups
Moseley	1913	Atomic Number	18 groups

TRICK - "1829-1864-1869-1913 = 3-8-9-18"

Triads3 → Octaves8 → Mendeleev9 groups → Modern18 groups.

16. Final Exam Traps — 12 Red Flags

- Dobereiner = 1829 triads; middle mass ≈ average of outer two
- Newland = 1864; every 8th; only up to Calcium
- Mendeleev = 1869, atomic mass, 9 groups, 63 elements
- Moseley = 1913, atomic number, 18 groups, 118 elements
- Eka-Aluminium = Gallium; Eka-Silicon = Germanium
- Group 17 = Halogens; Group 18 = Noble gases
- Noble gases are p-block
- Lanthanoids 58-71; Actinoids 90-103
- Period counts = 2-8-8-18-18-32-32
- Most electronegative = F; highest electron affinity = Cl
- Largest atom = Cs; source smallest non-noble = F
- Lightest metal = Li; highest MP = W

EXAM TRAP - "D-N-M-M / Eka / 17-18 / p / 58-71 90-103 / 2-8-8... / F-Cl / Cs-F / Li-W"

Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"History D-N-M-M = Dobereiner1829 triads3 middle=average → Newland1864 octaves every8th up to Ca → Mendeleev1869 atomic mass, 63 elements,7 periods,9 groups, no noble gas group, blank Eka spaces → Moseley1913 atomic number,118 elements,18 groups,7 periods → Eka B-A-S → Sc-Ga-Ge → Groups:1 Alkali,2 Alkaline,3-12 Transition,15 Pnictogen,16 Chalcogen,17 Halogen,18 Noble → f-block Lantha58-71 period6 + Acti90-103 period7; period count 2-8-8-18-18-32-32; blocks s1-2,p13-18,d3-12,f inner; H is s-block Group1 but non-metal → Trends: down Size+Metal↑ while EN/EA/IE↓; across opposite → F highest electronegativity, Cl highest electron affinity, Cs largest, source smallest F; Noble gases zero valency + high IE → Specials H smallest element, Cs largest, Li lightest metal, W highest MP, Mendeleev father."

TOP 5: 1829-1864-1869-1913 | Mendeleev mass/9 vs Moseley number/18 | Eka B-A-S→Sc-Ga-Ge | s-p-d-f group ranges | F electronegativity vs Cl electron affinity.

CHEMICAL BONDS & REACTIONS — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 4 · Bond Types + Sigma/Pi + Shapes + Oxidation No. + Reactions + Redox + Catalysts

Ye full notes nahi hai — Chapter 4 ke source mindmap se bond properties, sigma/pi counts, molecular shapes, oxidation-number rules, reaction types, redox logic aur catalyst-process pairs ko fast exam-recall tricks mein compress kiya gaya hai.

1. Chemical Bond — Stability + Octet

- Chemical bond = force of attraction between two or more atoms
- Bond formation ka source reason: stability achieve karna, outermost orbit mein 8 electrons complete karna, potential energy reduce karna
- Source lists 4 major bond types: Ionic, Covalent, Hydrogen, Coordinate

TRICK - "Bond = Attraction + 8 + Stability"
Outer shell octet complete → inert-gas type stability.

2. Four Bond Types — I-C-H-Co

Bond	Formation	Examples	Key Property
Ionic	Electron transfer	NaCl, MgO	Water soluble; solid bad conductor, aqueous/molten good
Covalent	Electron sharing	O2, H2O	Usually water-insoluble; poor conductor
Hydrogen	H with N/O/F	H2O, NH3, DNA	Water high BP; DNA H-bonds
Coordinate	Both shared e- from same atom	H2SO4, H3O+	Dative bond

TRICK - "I Transfer; C Share; H = N-O-F; Coordinate = Same Donor"
I-C-H-Co ka formation code.

EXAM TRAP - "Ionic: Solid BAD, Solution GOOD"
Aqueous/molten ionic compound conducts; solid does not.

3. Ionic Bond — Cation + Anion

- One atom loses electron → cation (+); other gains electron → anion (-)
- Na gives 1e- and Cl takes 1e- → NaCl
- Mg gives 2e- and O takes 2e- → MgO
- High melting point; soluble in water

TRICK - "Lose = Plus; Gain = Minus"
Electron loss → cation; electron gain → anion.

4. Covalent Bond — Share + Exceptions

- Usually non-metals share electron pairs
- O2 = non-polar covalent (same atoms)
- H2O = polar covalent (different atoms)
- Low melting/boiling point in source, exception Diamond
- Poor conductor in source, exception Graphite
- Soluble in organic solvents

EXAM TRAP - "Covalent Exceptions = Diamond HIGH MP + Graphite CONDUCTS"
D-G = two must-remember exceptions.

5. Hydrogen Bond — N-O-F with H

Type	Where	Examples / Effect
Intermolecular	Different molecules	H_2O, NH_3; high BP of water
Intramolecular	Same molecule	Protein, DNA, RNA, cellulose — as listed in source

TRICK - "H-Bond = H with N-O-F"
Source trap: NOT Cl or S.

EXAM TRAP - "Water High BP + DNA Double Helix = H-Bond"
High-frequency source facts.

6. Coordinate + Van der Waals

- Coordinate covalent/dative bond: shared pair ke dono electrons same donor atom se aate hain
- Source examples: H_2SO_4 and hydronium H_3O^+
- Van der Waals = weak intermolecular attraction/repulsion; chemical bonds se weaker
- Noble gases ke liquid/solid-state properties mein role

TRICK - "Coordinate = One Atom Gives BOTH"
H_3O^+ is a source example.

7. Sigma-Pi — 1 / 1+1 / 1+2

Bond	Composition	Pi Count	Examples
Single	1 sigma	0	H_2, CH_4, H_2O
Double	1 sigma + 1 pi	1	O_2, CO_2, C_2H_4
Triple	1 sigma + 2 pi	2	N_2, C_2H_2

TRICK - "Single 10; Double 11; Triple 12"
Read as sigma-pi counts: $1\sigma0\pi$ / $1\sigma1\pi$ / $1\sigma2\pi$.

EXAM TRAP - "Every Bond has Sigma; Pi only Double/Triple"
Triple bond = $1\sigma + 2\pi$, NOT 3σ.

8. Molecular Shapes — W-A-M-C

Molecule	Shape	Polarity
H_2O	Angular / V-shaped	Polar
NH_3	Trigonal pyramidal	Polar
CH_4	Tetrahedral	Non-polar
CO_2	Linear	Non-polar

TRICK - "Water V; Ammonia Pyramid; Methane Tetra; CO2 Line"
W-A-M-C shape chain.

EXAM TRAP - "H2O Polar; CO2 Non-polar despite polar bonds"
Source reason: CO_2 symmetry.

9. Valency — Fixed vs Variable

- Valency = combining ability; electrons used/shared/transferred
- Fluorine valency always -1; never positive in source
- Groups IA and IIA: valency = group number (Frankland rule)

- Cl: 2,8,7 → -1; Na: 2,8,1 → +1
- Fe, Cu, S, Hg, Sn = variable valency

TRICK - “F = Always -1”
Most electronegative; source says never positive.

TRICK - “Cross-Multiply Valencies → Formula”
Al(+3), Cl(-1) → $AlCl_3$.

10. Oxidation Number — H-O-F Rules

Species	Usual ON	Important Exception
H	+1	-1 in metal hydrides: NaH, CaH_2
O	-2	-1 in H_2O_2; +2 in OF_2; 0 in O_2
F	-1	Always -1
Na, K	+1	Always +1
Mg, Ca, Sr	+2	Always +2

- Neutral molecule: sum of oxidation numbers = 0
- Ion: sum of oxidation numbers = ion charge
- Oxidation number can be positive, negative, zero or fractional

TRICK - “H +1, O -2, F -1”
Exceptions: H metal hydride -1; O peroxide -1 / OF_2 +2.

EXAM TRAP -
F always -1; O is NOT always -2.

11. Nine Reaction Types — C-D-D-D-E-E-R-I-R

Type	Pattern / Source Hook
Combination	A + B → C
Decomposition	AB → A + B; $CaCO_3$ → CaO + CO_2
Displacement	AB + C → AC + B; more reactive displaces less
Double Displacement	AB + CD → AD + CB; ion exchange / precipitate
Exothermic	Energy released; combustion, respiration
Endothermic	Energy absorbed; digestion, photosynthesis, photochemical
Reversible	Both directions; equilibrium
Irreversible	One direction; completion
Redox	Oxidation + reduction simultaneously

TRICK - “C-D-D-D-E-E-R-I-R = 9 Reaction Types”
Combination, Decomposition, Displacement, Double-displacement, Exo, Endo, Reversible, Irreversible, Redox.

EXAM TRAP - “Respiration EXO; Photosynthesis ENDO”
Energy released vs absorbed.

12. Redox — OIL RIG

Process	Electron	Oxidation No.	Other Source Clues
Oxidation	Loss	Increases	Gain O/electronegative; lose H/electropositive
Reduction	Gain	Decreases	Gain H/electropositive; lose O/electronegative

TRICK - “OIL RIG”
Oxidation Is Loss; Reduction Is Gain (of electrons).

EXAM TRAP - "Oxidising Agent gets REDUCED; Reducing Agent gets OXIDISED"
Agent apne naam ka kaam dusre par karata hai.

13. Redox Example + Agents

- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- Fe: 0 → +2 = oxidised
- Cu^{2+}: +2 → 0 = reduced

Agent Type	Source Examples
Oxidising agents	H_2O_2, O_2, F_2, $KMnO_4$
Reducing agents	Li, Na, $NaBH_4$, H_2

EXAM TRAP - "H_2O_2 = BOTH"
Source says H_2O_2 can act as both oxidising and reducing agent.

14. Catalyst — Berzelius + Same/Different State

- Catalyst increases reaction rate without being consumed
- Discoverer in source = Berzelius
- Homogeneous catalysis = catalyst + reactants same physical state
- Heterogeneous catalysis = different physical states
- Positive catalyst increases rate; negative catalyst decreases rate

TRICK - "Catalyst = Rate Change, Not Consumed; Berzelius"
Homogeneous SAME state; Heterogeneous DIFFERENT state.

15. Process-Catalyst Master Table

Process	Catalyst / Source Note
Ghee from vegetable oil	Nickel (Ni)
Contact method - H_2SO_4	Pt powder, V_2O_5; source highlights V_2O_5 as most asked
Deacon process - Cl_2	Cupric chloride ($CuCl_2$)
Protein → Amino acids	Erepsin enzyme
Vinegar from sugar	Mycoderma aceti
Milk → Curd	Lactobacilli (Lactose)
Haber process - NH_3	Fe catalyst + Mo promoter

TRICK - "Ghee-Ni / Contact-V_2O_5 / Curd-Lacto / Haber-Fe+Mo"
Four highest-frequency catalyst pairs.

EXAM TRAP - "Haber ≠ Contact"
Haber = Fe + Mo; Contact H_2SO_4 = V_2O_5.

16. Final Exam Traps — 12 Red Flags

- Ionic solid = bad conductor; aqueous/molten = good conductor
- Diamond = covalent but hard/high MP; Graphite = covalent but conducts
- H-bond only source pair with N/O/F and H
- DNA double helix held by H-bonds
- Triple bond = 1σ + 2π, not 3σ
- H_2O V-shaped; CO_2 linear
- F oxidation number always -1; O usually -2 but -1 in H_2O_2
- Respiration exothermic; photosynthesis endothermic
- Displacement: more reactive displaces less reactive
- Oxidation = electron loss + ON increase; Reduction = electron gain + ON decrease
- Oxidising agent gets reduced; reducing agent gets oxidised
- Haber = Fe + Mo; Contact process = V_2O_5

EXAM TRAP - "Ionic-D/G-Hbond-Triple-Shapes-F/O-Exo/Endo-Displace-OILRIG-Agents-Haber/Contact"
Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Bond = attraction + octet stability → I-C-H-Co: Ionic transfer NaCl/MgO solid bad but solution good; Covalent share O_2/H_2O with Diamond high-MP + Graphite conductor exceptions; H-bond H with N-O-F, water high BP + DNA; Coordinate same atom gives both electrons, H_3O^+/H_2SO_4 → Sigma/Pi: single1σ, double1σ1π, triple1σ2π → Shapes H_2O V polar, NH_3 pyramidal polar, CH_4 tetra non-polar, CO_2 linear non-polar → Valency F always-1, variable Fe/Cu/S/Hg/Sn; ON H+1 except hydrides-1, O-2 except peroxide-1/OF_2+2, F-1 always; neutral sum0, ion sum=charge → Reactions 9: C-D-D-D-E-E-R-I-R; Respiration exo, Photosynthesis endo; displacement more reactive kicks less; double-displacement ion exchange/precipitate → Redox OIL RIG: oxidation loss e-/ON↑, reduction gain e-/ON↓; oxidiser gets reduced, reducer gets oxidised; Fe→Fe^{2+} oxidised, Cu^{2+}→Cu reduced; H_2O_2 both → Catalyst Berzelius; Homo same-state, Hetero different-state; Ghee-Ni, Contact-V_2O_5, Deacon-$CuCl_2$, Protein-Erepsin, Vinegar-Mycoderma aceti, Curd-Lactobacilli, Haber Fe+Mo."

TOP 5: Ionic solid bad/solution good | Diamond vs Graphite exceptions | Triple=1σ+2π | OIL RIG + agents reverse | Haber Fe+Mo vs Contact V_2O_5.

SOLUTION & COLLOID — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 5 · Solutions + Saturation + Colloid + Suspension + Tyndall + Emulsion + Buffer

Ye full notes nahi hai — Chapter 5 ke source mindmap se particle-size ranges, solution types, saturation-temperature effects, Solution–Colloid–Suspension differences, Tyndall effect, colloid applications, emulsions aur buffer facts ko fast exam-recall format mein compress kiya gaya hai.

1. Solution Basics — Less + More = Solution

- Solution = homogeneous mixture of two or more substances
- Solute = dissolved substance; usually less quantity
- Solvent = dissolving medium; usually more quantity
- Universal solvent = Water (H_2O)
- Source examples: solute = sugar/salt/NaCl; solvent = water/alcohol/acetone

TRICK - "Solute LESS + Solvent MORE = Solution"
Universal Solvent = Water.

2. Particle Size Master — <1 / 1–1000 / >1000 nm

System	Particle Size	Nature	Tyndall	Visible?	Separation
True Solution	< 1 nm	Homogeneous	No	No	Not by filtration
Colloid	1–1000 nm	Appears homogeneous; actually heterogeneous	Yes	No	Centrifugation
Suspension	> 1000 nm	Heterogeneous	Yes	Yes	Filtration

TRICK - "1 / 1000 Border Rule"
True Solution <1 nm → Colloid 1–1000 nm → Suspension >1000 nm.

EXAM TRAP - "Tyndall = Colloid + Suspension; NOT True Solution"
Visible to naked eye = Suspension only.

3. Solution Types — Solute in Solvent

Solution	Solute	Solvent	Source Example
Solid Solution	Gas	Solid	H_2 absorbed in Palladium
Solid Solution	Liquid	Solid	Mercury amalgam with Sodium
Solid Solution	Solid	Solid	Zn in Cu = Brass / alloys
Liquid Solution	Gas	Liquid	O_2 in water
Liquid Solution	Liquid	Liquid	Alcohol in water
Liquid Solution	Solid	Liquid	Sugar in water
Gaseous Solution	Gas	Gas	Air
Gaseous Solution	Liquid	Gas	Chloroform + N_2 = Fog
Gaseous Solution	Solid	Gas	Camphor + N_2 = Smoke

TRICK - "Pd-H_2 = Gas in Solid; Amalgam = Liquid in Solid; Air = Gas in Gas"
Three high-frequency type examples.

EXAM TRAP - "Fog = Liquid in Gas; Smoke = Solid in Gas"
O_2 in water = Gas in Liquid.

4. Saturated vs Unsaturated

Feature	Saturated	Unsaturated
At given T & P	Maximum solute dissolved	More solute can dissolve
Extra solute	Remains undissolved	Can still dissolve
Temperature ↑	Becomes unsaturated	Can dissolve more
Temperature ↓	Capacity falls	Can become saturated

TRICK - “Temp UP → Saturated becomes UNSATURATED”
More solute can dissolve after temperature increase.

EXAM TRAP - “Temp DOWN → Unsaturated can become SATURATED”
Solubility capacity decreases.

- Most solids: solubility increases with temperature
- Gases in liquids: solubility decreases with temperature
- Source example: carbonated drinks go flat when warm; O_2 solubility in water falls as temperature rises

5. Solid vs Gas Solubility — Opposite Trend

Solute Type	Temperature ↑
Most solids in liquid	Solubility ↑
Gases in liquid	Solubility ↓

EXAM TRAP - “Solid Likes Heat; Gas Escapes Heat”
Most solids ↑ solubility; dissolved gases ↓ solubility.

6. True Solution vs Colloid vs Suspension

Feature	True Solution	Colloid	Suspension
Nature	Homogeneous	Pseudo-homogeneous / actually heterogeneous	Heterogeneous
Particle	<1 nm	1–1000 nm	>1000 nm
Tyndall	No	Yes	Yes
Naked eye	No	No	Yes
Separation	Not by filtration	Centrifugation	Filtration
Examples	Sugar/salt water	Milk, Ink, Blood, Fog, Smoke	Chalk/Mud in water

TRICK - “Solution-COLLOID-Suspension = No / CENTRI / FILTER”
True solution not by filtration; colloid centrifugation; suspension filtration.

EXAM TRAP - “Milk = Colloid, NOT True Solution”
Milk separated by centrifugation to obtain butter/cream.

7. Tyndall Effect — Light Scattering

- Tyndall effect = scattering of a beam of light by colloidal particles
- Observed in Colloid + Suspension; NOT in True Solution
- Source examples: sunlight beam in dusty room; headlights in fog
- Source states blue colour of sky = Tyndall effect because shorter-wavelength blue light scatters more

TRICK - “Tyndall = Light SCATTER by Colloid”
Headlights + Fog; Dusty-room sunlight.

EXAM TRAP -
Source framing: Blue colour of sky = Tyndall Effect, not reflection.

8. Colloid Applications — S-C-B-R-D

Application	Source Recall
Smoke precipitation	Removing smoke from chimneys
Cleaning clothes	Detergent action
Blood clotting	$FeCl_3$ as coagulant
Rain formation	Cloud colloids coagulate around dust
Delta formation	River silt colloid coagulated by sea salt

TRICK - “S-C-B-R-D = Smoke-Cleaning-Blood-Rain-Delta”
Colloid applications chain.

TRICK - “Delta = Sea Salt Coagulates River Silt”
River-sea junction exam point.

EXAM TRAP - “Blood + $FeCl_3$ = Coagulation”
Source says $FeCl_3$ can stop bleeding.

9. Emulsion — Two Immiscible Liquids

- Emulsion = colloidal system of two immiscible liquids
- All emulsions are colloids; source particle range 1–1000 nm

Type	Meaning	Examples
O/W	Oil droplets in water	Milk, Mayonnaise, Vanishing cream
W/O	Water droplets in oil	Butter, Cold cream, Margarine

TRICK - “Milk = O/W; Butter = W/O”
Mayonnaise O/W; Cold Cream W/O.

EXAM TRAP - “O/W = Oil is Inside Water; W/O = Water is Inside Oil”
Slash ke left wala dispersed phase yaad rakho.

10. Buffer Solution — pH Stable

- Buffer = solution whose pH does not change significantly after small acid/base addition
- Resists change in pH
- Source example: blood pH ≈ 7.4 maintained by buffer
- Uses: medicine, food preservation, chemical labs

TRICK - “Buffer = pH Bodyguard”
Small acid/base addition par pH ko stable rakhta hai.

EXAM TRAP - “Blood ≈ 7.4”
Biological buffer number hook.

11. Quick Example Hooks

Cue	Answer
Universal solvent	Water
H_2 in Pd	Gas in Solid
Mercury amalgam	Liquid in Solid
Air	Gas in Gas
Fog	Liquid in Gas
Smoke	Solid in Gas
O_2 in water	Gas in Liquid
Milk	Colloid + O/W emulsion
Butter	W/O emulsion
Colloid separation	Centrifugation
Suspension separation	Filtration
Blood pH buffer	≈ 7.4

12. Final Exam Traps — 12 Red Flags

- True solution <1 nm; Colloid 1–1000 nm; Suspension >1000 nm
- Tyndall effect = Colloid + Suspension; not True Solution
- Suspension particles visible; colloid particles not visible to naked eye
- Colloid appears homogeneous but is actually heterogeneous
- Colloid → centrifugation; Suspension → filtration
- Milk = colloid, not true solution
- Milk = O/W; Butter = W/O
- Temp ↑ → saturated solution becomes unsaturated
- Gas solubility in liquid decreases with temperature
- Fog = liquid in gas; Smoke = solid in gas
- Delta formation = colloidal coagulation by sea salt
- Buffer resists pH change; blood pH source ≈ 7.4

EXAM TRAP - "Size-Tyndall-Visible-Hetero-Centri/Filter-Milk-O/W-Temp-Gas-Fog/Smoke-Delta-Buffer"
Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Solution = Solute LESS + Solvent MORE; water universal → particle size: True<1nm, Colloid1–1000nm, Suspension>1000nm; Tyndall Colloid+Suspension only, naked-eye Suspension only → Solution types: H_2/Pd gas-solid, amalgam liquid-solid, Brass solid-solid, O_2/water gas-liquid, alcohol/water liquid-liquid, sugar/water solid-liquid, Air gas-gas, Fog liquid-gas, Smoke solid-gas → Saturated max solute at given T/P; Temp↑ makes it unsaturated; Temp↓ can saturate; most solids solubility↑ with heat but gas-in-liquid solubility↓ → True solution homogeneous/no Tyndall/not by filtration; Colloid pseudo-homogeneous actually heterogeneous/Tyndall/Centrifugation; Suspension heterogeneous/Tyndall/visible/Filtration → Tyndall = scattering; source links blue sky, fog headlights, dusty-room beam → Colloid apps S-C-B-R-D: smoke, clothes, blood $FeCl_3$, rain, delta coagulation → Emulsion = two immiscible liquids: Milk/Mayonnaise/Vanishing O/W; Butter/Cold Cream/Margarine W/O → Buffer = pH bodyguard; blood≈7.4."

TOP 5: <1 / 1–1000 / >1000 nm | Tyndall: Colloid+Suspension | Colloid=Centrifugation vs Suspension=Filtration | Milk O/W vs Butter W/O | Gas solubility ↓ with temperature.

ACID, BASE & SALT — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 6 · Acid/Base Concepts + Aqua Regia + Acid Sources + Salts + Indicators + pH

Ye full notes nahi hai — Chapter 6 ke source mindmap se acid-base concepts, natural acid sources, base uses, salt formulas, indicator colours, pH values aur high-frequency exam traps ko fast recall tricks mein compress kiya gaya hai.

1. Acid vs Base — pH + Taste + Litmus

Property	Acid	Base
pH	< 7	> 7
Taste	Sour	Bitter
Litmus	Blue → Red	Red → Blue
Arrhenius	Gives H+	Gives OH-
Brønsted	Proton donor	Proton acceptor
Lewis	Accepts e- pair	Donates e- pair

TRICK - "Acid = H+ + Sour + Blue→Red"
Base = OH- + Bitter + Red→Blue.

EXAM TRAP - "Lewis = Acid Accepts, Base Donates"
Lewis rule Arrhenius/Bronsted se ulta-feel deta hai, isliye alag yaad karo.

2. Acid Concepts — A-B-L

Concept	Acid Definition	Source Example
Arrhenius	Releases H+ in aqueous solution	HCl, H2SO4
Brønsted-Lowry	Proton donor	HCl, H2SO4
Lewis	Electron-pair acceptor	BF3, AlCl3

TRICK - "A-B-L = H+ / Proton / e-Pair Accept"
Arrhenius → H+ in water.
Brønsted → proton donor.
Lewis → electron-pair acceptor.

3. Acid Reactions — M-C-B

Reaction	Pattern	Example
With Metal	Acid + Metal → Salt + H2	Zn + 2HCl → ZnCl2 + H2
With Carbonate	Acid + Carbonate → Salt + H2O + CO2	CaCO3 + H2SO4 → CaSO4 + H2O + CO2
With Base	Acid + Base → Salt + Water	HCl + KOH → KCl + H2O

TRICK - "M-C-B = H2 / CO2 / Neutralisation"
Metal gives H2; carbonate gives CO2; base gives salt+water.

4. Aqua Regia — Royal 3:1

- Aqua Regia = fresh mixture of 3 parts concentrated HCl + 1 part concentrated HNO3
- Called "Royal Water"
- Dissolves Gold (Au) and Platinum (Pt)
- Must be freshly prepared

TRICK - "Aqua Regia = 3 HCl : 1 HNO3"
Gold + Platinum dissolve.

EXAM TRAP -
Ratio 3:1 HCl:HNO3 — NOT 1:3.

5. Special Acids — HCl / HF / Benzoic / Glacial

Acid	Source Special Fact
HCl	Helps digestion; stomach acid
HF	Dissolves glass
Benzoic acid	Used in medicine
Glacial acetic acid	99–100% pure acetic/ethanoic acid

TRICK - “HCl Stomach; HF Glass; Glacial = Pure Acetic”
HF is the glass-dissolving trap.

EXAM TRAP -
Glass dissolves with HF, not HCl.

6. Natural Acid Sources — F-C-O-A

Acid	Natural Source	Memory Hook
Formic	Red ants, scorpions	Ant sting
Citric	Lemon, orange, tomato	Citrus
Oxalic	Spinach, beetroot	Leafy/Beet
Acetic	Vinegar	Vinegar acid

TRICK - “F-C-O-A = Ant - Citrus - Spinach - Vinegar”
Formic→ants/scorpions; Citric→lemon/orange/tomato; Oxalic→spinach/beetroot; Acetic→vinegar.

EXAM TRAP - “Contact H_2SO_4 Catalyst = V_2O_5”
Source high-frequency acid-production fact.

7. Bases — Arrhenius / Brønsted / Lewis

Concept	Base Definition	Source Example
Arrhenius	Gives OH^- in water	NaOH, KOH
Brønsted-Lowry	Accepts H^+	NH_3, H_2O
Lewis	Donates electron pair	Cl^-, F^-, OH^-

TRICK - “Base = OH^- / Proton Accept / e-Pair Donate”
Acid-base concept mirror.

- Base taste = bitter
- Red litmus → Blue
- pH > 7
- Source note: insoluble bases ZnO, FeO react with acids but do not have basic properties in water

8. Base Uses — M-N-C-K

Base	Common Name	Use
$Mg(OH)_2$	Milk of Magnesia	Stomach antacid
NaOH	Caustic Soda	Soap making, petroleum refining
$Ca(OH)_2$	Slaked Lime	Treating acid burns
KOH	Caustic Potash	Lab reagent; absorbs CO_2 & SO_2

TRICK - “M-N-C-K = Magnesia-Stomach / NaOH-Soap / CaOH-Acid burn / KOH-Gas absorb”
Four source exam-use pairs.

EXAM TRAP -
Stomach acidity = $Mg(OH)_2$, NOT NaOH.

9. Salt — Neutralisation Product

- Salt = product of neutralisation reaction of acid + base
- Acid + Base → Salt + Water
- Source example: $HNO_3 + KOH \rightarrow KNO_3 + H_2O$

Salt Type	Description	Example
Common Salt	Sodium chloride	$NaCl$
Acidic Salt	Partial neutralisation of polyprotic acid	$NaHCO_3$
Basic Salt	Partial neutralisation of weak base with strong acid	$Mg(OH)Cl$
Double Salt	Mixing two or more salts	Alum
Mixed Salt	One acid+two bases OR two acids+one base	$CaOCl_2$

TRICK - "Acidic-NaHCO3; Basic-Mg(OH)Cl; Double-Alum; Mixed-Bleaching"
Salt-type mapping.

EXAM TRAP -
Bleaching powder = Mixed salt, NOT Double salt.

10. Important Salts — B-W-B-P

Salt	Formula	Main Use
Baking Soda	$NaHCO_3$	Fire extinguisher + baking/yeast agent
Washing Soda	Na_2CO_3	Treats permanent hardness; glass/paper/soap
Bleaching Powder	$Ca(OCl)_2$	Pool cleaning + water disinfection + chloroform
Plaster of Paris	$CaSO_4 \cdot 1/2H_2O$	Broken bones, chalk, toys, decoration

TRICK - "B-W-B-P = Baking-Washing-Bleaching-POP"
$NaHCO_3$ / Na_2CO_3 / $Ca(OCl)_2$ / $CaSO_4 \cdot 1/2H_2O$.

EXAM TRAP - "POP Half-Water; Gypsum Two-Water"
POP = $CaSO_4 \cdot 1/2H_2O$; Gypsum = $CaSO_4 \cdot 2H_2O$.

11. Crystalline Salt Formulas — E-M-A-B-W

Name	Formula
Epsom Salt	$MgSO_4 \cdot 7H_2O$
Mohr's Salt	$FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
Alum	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$
Blue Vitriol	$CuSO_4 \cdot 5H_2O$
White Vitriol	$ZnSO_4 \cdot 7H_2O$

TRICK - "E-M-A-B-W = 7-6-24-5-7 waters"
Epsom7; Mohr6; Alum24; Blue5; White7.

EXAM TRAP - "Blue Vitriol = $CuSO_4 \cdot 5H_2O$; Epsom = $MgSO_4 \cdot 7H_2O$"
Two most asked hydrate formulas.

12. Indicators — Colour Master

Indicator	Acid	Base
Litmus	Red	Blue
Phenolphthalein	Colourless	Pink
Methyl Orange	Red	Yellow
Methyl Red	Red	Yellow
Phenol Red	Yellow	Red/Pink

TRICK - “Phenolphthalein = C→P; Methyl Orange = R→Y; Phenol Red = Y→R”
Acid→Base colour order.

EXAM TRAP - “Phenol Red is OPPOSITE of Methyl Orange”
Methyl Orange acid red/base yellow; Phenol Red acid yellow/base red/pink.

13. Olfactory Indicators — Smell Rule

- Clove oil, Vanilla, Onion = olfactory indicators
- In Base: smell lost
- In Acid: smell retained

EXAM TRAP - “Base = Smell BYE; Acid = Aroma Stays”
Clove-Vanilla-Onion same rule.

14. pH Scale — Sørensen + 0–14

- pH = measure of H^+ concentration
- Discovered by Sørensen
- $pH = -\log[H^+]$
- $[H^+] = 10^{(-pH)}$
- Acidic = 0–6; Neutral = 7; Basic = 8–14
- Higher pH = more basic; lower pH = more acidic

TRICK - “Sørensen = pH; 7 = Neutral”
0 strongest acid side; 14 strongest base side.

EXAM TRAP -
pH concept = Sørensen, NOT Arrhenius or Brønsted.

15. pH Values — Number Hooks

Substance	Source pH
Lemon Juice	2.0–2.5
Vinegar	2.4–3.4
Milk	6.4–6.8
Saliva	6.2–7.6
Pure Water	7.0
Tears	7.0–7.5
Blood Plasma	7.35–7.45
Sea Water	7.8–8.4

TRICK - “Lemon 2; Water 7; Blood 7.4; Sea 8”
Quick pH ladder.

EXAM TRAP - “Blood 7.35–7.45; ±0.2 change = Death”
Source-specific exam statement.

16. Final Exam Traps — 12 Red Flags

- Lewis Acid accepts e- pair; Lewis Base donates e- pair
- Aqua Regia = 3:1 $HCl:HNO_3$, not 1:3
- HF dissolves glass; HCl is stomach acid
- Glacial acetic acid = 99–100% pure acetic acid
- Formic acid = red ants/scorpions; Oxalic = spinach/beetroot
- $Mg(OH)_2$ = stomach antacid; NaOH = soap making

- Bleaching powder = Mixed salt, not double salt
- Washing Soda = Na_2CO_3; Baking Soda = $NaHCO_3$
- POP = $CaSO_4 \cdot 1/2H_2O$; Gypsum = $CaSO_4 \cdot 2H_2O$
- Phenolphthalein acid = colourless, base = pink
- Olfactory indicators lose smell in base, retain in acid
- pH discovered by Sørensen; blood source pH 7.35–7.45

EXAM TRAP - "Lewis-Aqua-HF-Glacial-AcidSources-Bases-Mixed-Wash/Bake-POP-Ind-Olfactory-pH"
Chapter ke 12 highest-yield traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Acid pH<7 sour blue→red; Base pH>7 bitter red→blue; Arrhenius H^+/OH^-, Brønsted proton donor/acceptor, Lewis acid accepts e-pair/base donates → Acid reactions: metal→H_2, carbonate→CO_2, base→salt+water → Aqua Regia fresh 3HCl:1HNO_3 dissolves Au/Pt; HCl stomach, HF glass, glacial acetic 99–100% → Acid sources F-C-O-A = ants/scorpions, citrus, spinach/beetroot, vinegar; H_2SO_4 Contact catalyst V_2O_5 → Base uses: $Mg(OH)_2$ antacid, NaOH soap, $Ca(OH)_2$ acid burns, KOH absorbs CO_2/SO_2 → Salt = neutralisation; acidic $NaHCO_3$, basic Mg(OH)Cl, double Alum, mixed $CaOCl_2$; Baking $NaHCO_3$, Washing Na_2CO_3, Bleaching $Ca(OCl)_2$, POP $CaSO_4 \cdot 1/2H_2O$ → Hydrates E-M-A-B-W = Epsom7, Mohr6, Alum24, Blue5, White7 → Indicators: Phenolphthalein C/P, Methyl Orange R/Y, Phenol Red Y/R; olfactory smell lost in Base → pH Sørensen, -log[H^+], acid0–6 neutral7 base8–14; Lemon≈2, Water7, Blood7.35–7.45, Sea7.8–8.4."

TOP 5: Aqua Regia 3:1 | HF dissolves glass | Baking $NaHCO_3$ vs Washing Na_2CO_3 | POP half-water vs Gypsum two-water | Phenolphthalein colourless→pink + Sørensen pH.

METALS, NON-METALS & METALLURGY — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 7 · Reactivity + Ores + Non-metals + Alloys + Water/H2O2 + Metallurgy

Ye full notes nahi hai — Chapter 7 ke source mindmap se metal facts, reactivity series, ores, non-metals/metalloids, alloys, hydrogen-water-heavy water/H2O2 aur metallurgy steps ko fast exam-recall tricks mein compress kiya gaya hai.

1. Metal vs Non-metal — Lose vs Gain

Type	Electron Behaviour	Ion	Oxide Nature
Metal	Lose electrons	Cation (+)	Usually basic / alkaline
Non-metal	Accept electrons	Anion (-)	Usually acidic

TRICK - "Metal LOSES → + ; Non-metal GAINS → -"
Source exception: Hydrogen can behave as both metal and non-metal.

2. Metal Superlatives — Ag-Au-Al-Hg-Li-Os-Ti

Category	Source Answer
Best conductor	Silver (Ag)
Most ductile	Gold (Au); then Silver
Most abundant metal in Earth crust	Aluminium (Al) ≈ 8%
Liquid metal	Mercury (Hg)
Lightest metal / strongest reducing agent	Lithium (Li)
Heaviest metal	Osmium (Os)
Strategic metal	Titanium (Ti)
Radioactive liquid metal — source note	Francium (Fr)

TRICK - "Ag Conducts, Au Ductile, Al Abundant, Hg Liquid, Li Light, Os Heavy, Ti Strategic"
One-line metal-superlative chain.

EXAM TRAP - "Wilson = Copper; Minamata = Mercury"
Excess Cu → Wilson's disease; excess Hg → Minamata disease.

3. Reactivity Series — K Na Ca Mg Al Fe Pb H Cu Hg Ag Au

- Most → least reactive: K > Na > Ca > Mg > Al > Fe > Pb > H > Cu > Hg > Ag > Au
- More reactive metal displaces less reactive metal from solution
- Example: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- H is placed between Pb and Cu

TRICK - "King NaCa Maggi, Al Fe Pb; H Cu-Hg Ag-Au"
K-Na-Ca-Mg-Al-Fe-Pb-H-Cu-Hg-Ag-Au.

EXAM TRAP - "Above H = More Active; Below H = Less Active"
Source series placement trap.

4. Reactivity → Extraction Method

Activity	Metals	Method
Highly active	K, Na, Ca, Mg, Al	Electrolysis in molten state
Medium: carbonate ores	Zn, Fe, Pb, Cu	Calcination → oxide → reduction/smelting
Medium: sulphide ores	Zn, Fe, Pb, Cu	Roasting → oxide → reduction
Less active	Ag, Au	Simple heating / native-free state

TRICK - "High = ELECTRO; Carbonate = CALCINE; Sulphide = ROAST; Low = FREE"
Extraction method directly reactivity/ore type se jodo.

5. Metal Properties + Amphoteric

- Metals generally hard, shiny, malleable and ductile
- Good conductors of heat/electricity: source highest Ag, lowest Pb
- Source non-shiny exceptions: Mercury, Caesium, Gallium
- Metal oxides generally basic
- Amphoteric oxides/metals in source: Al, Zn, Pb — react with both acid and base

EXAM TRAP - "Amphoteric = A-Z-P"
Aluminium - Zinc - Lead.

6. Flame Colours — Na-K-Rb-Li-Ba-Sr

Metal	Flame Colour
Na	Golden Yellow
K	Purple / Lilac
Rb	Red Violet
Li	Crimson Red
Ba	Apple Green
Sr	Red

TRICK - "Na Yellow; K Lilac; Ba Apple; Li Crimson; Sr Red"
Firework/street-lamp colour traps.

7. Important Metal Uses

Metal	Source Use
Na	Street lamps — yellow light
Cu	Wires, utensils, foil
Al	Wire, space industry
Hg	Thermometer, tube lights
Mg	Flash bulb making
Fe	Steel industry
Cd	Moderator in nuclear reactors — source note
Ba / Sr	Green / red fireworks

TRICK - "Na Street; Mg Flash; Fe Steel; Ba Green; Sr Red"
Quick use-memory chain.

8. Ores — Main Pair Master

Metal	Main / Important Ore	Formula
Al	Bauxite; Cryolite/Corundum	$Al_2O_3 \cdot 2H_2O$; Na_3AlF_6 / Al_2O_3
Cu	Chalcopyrite	$CuFeS_2$
Fe	Haematite; Magnetite	Fe_2O_3; Fe_3O_4
Pb	Galena	PbS
Zn	Zincblende; Calamine	ZnS; $ZnCO_3$
Mn	Pyrolusite	MnO_2
Ag	Argentite	Ag_2S
Ca	Calcite	$CaCO_3$
Mg	Magnesite; Dolomite	$MgCO_3$; $CaMg(CO_3)_2$
Cd	Greenockite	CdS

TRICK - "Al-Bauxite, Cu-Chalco, Fe-Haema, Pb-Galena, Mn-Pyro"
Five highest-frequency ore pairs.

EXAM TRAP - "All Ores are Minerals; NOT all Minerals are Ores"
Ore must allow easy and economical metal extraction.

9. Non-metals — 22 = 11 Gas + 1 Liquid + 10 Solid

- Source total non-metallic elements = 22: 11 gases, 1 liquid, 10 solids
- Mostly bad conductors; Graphite = good conductor exception
- Diamond = hardest non-metal / very high melting point
- Bromine = only liquid non-metal at room temperature
- Non-metal oxides generally acidic
- Hydrogen behaves as both metal and non-metal — source exception

TRICK - "22 = 11G + 1L + 10S"
Liquid non-metal = Bromine.

EXAM TRAP - "Graphite Conducts; Diamond Hardest"
Two classic carbon exceptions.

10. Metalloids — 8 in p-block

- Have both metallic and non-metallic properties
- Present only in p-block — source note
- Total metalloids = 8
- Examples: B, Si, Ge, As, Sb, Te, Po, At
- Source note: Boron fibre used in bulletproof jackets / lightweight aircraft material

TRICK - "B-Si-Ge-As-Sb-Te-Po-At = 8 Metalloids"
All source-listed in p-block.

11. Noble Gases — He Ne Ar Kr Xe Rn

Gas	Source Use
He	Balloons/light aircraft; divers' artificial respiration; nuclear-reactor coolant
Ar	Welding; incandescent bulbs; tube lights with Hg vapour
Ne	Advertising neon signs
Xe	Photography flash tubes; "Stranger gas"

TRICK - "He-Ne-Ar-Kr-Xe-Rn = 6 Group-18 Gases"
Complete outer shell = chemically inert in source framing.

EXAM TRAP - "He Balloons; Ar Welding/Bulb; Ne Signs; Xe Flash"
Use-based recall.

12. Alloys — Brass/Bronze/German Silver/Solder/Steel

Alloy	Composition	Use / Trap
Brass	Cu 70% + Zn 30%	Utensils, wires, pipes
Bronze	Cu 80% + Sn 20%	Statues; source calls oldest alloy
German Silver	Cu 50% + Zn 30% + Ni 20%	Jewellery; NO silver
Solder	Sn 50% + Pb 50%	Low MP; joining/plating
Stainless Steel	Fe 70% + Cr 18% + Ni 8% + C 1%	Cr gives rust resistance
Amalgam	Any metal + Hg	Dental fillings; Fe & Pt do NOT form amalgam

TRICK - "Brass = Cu+Zn; Bronze = Cu+Sn"
Zn vs Sn is the classic swap.

EXAM TRAP - "German Silver = NO Ag; Amalgam = Hg, but Fe/Pt NO"
Stainless Steel rust resistance comes from Chromium.

13. Hydrogen — Cavendish 1766

- Hydrogen discovered by Henry Cavendish — 1766
- Atomic mass = 1.008; electronic configuration = $1s^1$
- One proton + one electron; source says no neutron
- Diatomic H_2
- Source: 70% of Universe by mass; 0.15% of Earth atmosphere; 3rd most abundant on Earth
- Uses: rocket fuel, NH_3 synthesis, fuel cell

TRICK - "H = Cavendish 1766 + 1p 1e 0n + H2"
Universe source figure = 70% by mass.

14. Water — 4°C + 104.5°

Fact	Source Recall
Universal solvent	H_2O
pH	7 — neutral pure water
Maximum density / minimum volume	4°C
H-O-H bond angle	104.5°
Density	Ice 0.917; Water 0.997; Gas 0.6

TRICK - "Water = 4°C + 104.5° + pH7"
Three-number memory chain.

EXAM TRAP - "Water Densest at 4°C"
Water densest at 4°C, NOT 0°C.

15. Hard vs Soft Water

Type	Cause / Behaviour	Removal
Temporary hardness	Ca/Mg bicarbonates; poor soap lather	Boiling or Clark's method
Permanent hardness	Ca/Mg sulphates & chlorides	Washing soda, Calgon, ion-exchange/resin
Soft water	Free from Ca/Mg salts; good lather	Rainwater naturally soft — source

EXAM TRAP - "Temporary = BOIL; Permanent = WASH"
Boiling/Clark vs Washing Soda/Calgon/Ion exchange.

16. Heavy Water + H2O2

Substance	Discoverer / Year	Source Hooks
Heavy Water D_2O	Harold Urey — 1932	Deuterium oxide; nuclear-reactor moderator; reaction mechanism; deuterium compounds
Hydrogen Peroxide H_2O_2	Thénard — 1818	"Oxygenated Water"; both oxidant & reducing agent; bleaching/antiseptic; rocket fuel

TRICK - "D2O = Urey 1932; H2O2 = Thénard 1818"
Heavy water = reactor moderator.

EXAM TRAP - "H2O2 = BOTH Oxidant + Reducer"
Source high-frequency redox trap.

17. Metallurgy — Ore to Refined Metal

- Metallurgy = extraction of metal from ore
- Source sequence: Ore → Concentration → Metal Oxide → Reduction/Smelting → Crude Metal → Refining

TRICK - “O-C-O-R-C-R”
Ore - Concentration - Oxide - Reduction - Crude - Refining.

18. Metallurgy Terms — Mineral/Ore/Flux/Slag

Term	Source Definition / Hook
Mineral	Naturally occurring inorganic solid; not all minerals are ores
Ore	Metal extracted easily & economically; all ores are minerals
Flux	Converts infusible impurities into fusible slag; CaO basic / SiO_2 acid flux
Slag	Fusible waste formed by flux + impurities

EXAM TRAP - “Flux + Gangue → Slag”
Flux makes impurity fusible/removable.

19. Calcination vs Roasting

Process	Air	Ore Type	Purpose
Calcination	Absent / limited air	Carbonate ores	Remove volatile impurities; form oxide
Roasting	Excess air	Sulphide ores	Oxidise ore/impurities; form oxide

TRICK - “CALCINATION = Carbonate + Cut Air”
ROASTING = Sulphide + lots of Oxygen/Air.

EXAM TRAP - “Calcination ≠ Roasting”
Calcination = no/limited air; Roasting = excess air.

20. Refining — Distillation vs Liquation

Method	Used For	Metals
Distillation	Low boiling-point metals	Zn, Cd, Hg
Liquation	Low melting-point metals	Sn, Hg, Pb

TRICK - “Distil = Z-C-H; Liquate = S-H-P”
Zn-Cd-Hg / Sn-Hg-Pb.

21. Final Exam Traps — 14 Red Flags

- Best conductor = Ag; most ductile = Au; most abundant crust metal = Al
- Liquid metal = Hg; liquid non-metal = Br
- Wilson = excess Cu; Minamata = excess Hg
- Reactivity series puts H between Pb and Cu
- K/Na/Ca/Mg/Al extraction = electrolysis
- Bauxite = Al main ore; Chalcopyrite = Cu; Haematite = Fe; Galena = Pb
- Graphite conducts; Diamond hardest non-metal
- Brass = Cu+Zn; Bronze = Cu+Sn; German Silver has NO Ag
- Fe and Pt do NOT form amalgam — source note
- Water max density at 4°C; bond angle 104.5°
- Temporary hardness = boiling; permanent = washing soda/Calgon/ion exchange
- D_2O = Urey1932; H_2O_2 = Thénard1818 and both oxidant/reducer
- Calcination = absent/limited air; Roasting = excess air
- Distillation: Zn/Cd/Hg; Liquation: Sn/Hg/Pb

EXAM TRAP - “Ag-Au-Al / Hg-Br / Cu-Hg / H-place / Electro / Ores / Graphite-Diamond / Alloys / Amalgam / 4°-104.5° / Hardness / D2O-H2O2 / Calc-Roast / Dist-Liq”
Chapter ke 14 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Metal loses e-→cation/basic oxide; non-metal gains e-→anion/acidic oxide; H exception → Ag best conductor, Au most ductile, Al8% crust, Hg liquid, Li light, Os heavy, Ti strategic; Wilson-Cu, Minamata-Hg → Reactivity K>Na>Ca>Mg>Al>Fe>Pb>H>Cu>Hg>Ag>Au; high active electrolysis, carbonate calcination, sulphide roasting, Ag/Au native → Flames Na yellow, K lilac, Ba apple-green, Li crimson, Sr red → Ores: Al-Bauxite, Cu-Chalco $CuFeS_2$, Fe-Haema Fe_2O_3/Magnetite Fe_3O_4, Pb-Galena PbS, Mn-Pyro MnO_2; all ores minerals not vice-versa → Non-metals source22=11G+1L+10S; Graphite conductor, Diamond hard, Bromine liquid; metalloids8 B-Si-Ge-As-Sb-Te-Po-At; noble gases HeNeArKrXeRn → Alloys: Brass CuZn, Bronze CuSn, German Silver CuZnNi no Ag, Solder SnPb, Stainless FeCrNiC with Cr rust resistance, amalgam=Hg but Fe/Pt no → H Cavendish1766, 1p1e0n; Water pH7, density max4°C, angle104.5°; temporary hardness boil, permanent wash/Calgon/ion exchange → D_2O Urey1932 reactor moderator; H_2O_2 Thénard1818 both oxidant+reducer → Metallurgy O-C-O-R-C-R; flux+gangue→slag; Calcination carbonate/no air vs Roasting sulphide/excess air; Distil ZnCdHg, Liquate SnHgPb."

TOP 5: Reactivity series + extraction | Al-Bauxite/Cu-Chalco/Fe-Haema/Pb-Galena | Brass vs Bronze vs German Silver | Water 4°C/104.5° + hardness | Calcination vs Roasting.

ORGANIC CHEMISTRY — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 8 · Carbon + Allotropes + Hydrocarbons + Nomenclature + Functional Groups + Fuels

Ye full notes nahi hai — Chapter 8 ke source mindmap se carbon facts, allotropes, hydrocarbon formulae, nomenclature prefixes, functional groups, important organic compounds, soap-detergent aur fuel-quality traps ko fast exam-recall tricks mein compress kiya gaya hai.

1. Carbon — 6 / 12.011 / +4 / Group 14

Property	Source Recall
Symbol	C
Atomic Number	6
Atomic Mass	12.011
Valency	+4
Electronic configuration	[He] $2s^2$ $2p^2$
Group / Period	14 / 2
Bond type	Covalent
Electrical conduction	Bad conductor; Graphite exception

TRICK - "Carbon = 6 - 12.011 - 4 - 14/2"
Atomic no. 6; mass 12.011; valency +4; Group 14 Period 2.

- Organic chemistry = study of carbon compounds, source excludes CO, CO_2 and carbonates
- Carbon forms 4 covalent bonds
- Catenation = self-chain forming ability → long, branched and ring structures
- Source says silicon has only limited catenation

EXAM TRAP - "Catenation = Carbon Chains with Carbon"
Long chains + branched chains + rings.

2. Carbon Isotopes — 12 / 13 / 14

Isotope	Status	Source Hook
C-12	Stable	98.9% natural
C-13	Stable	1.1% natural
C-14	Radioactive	Used in carbon dating

EXAM TRAP - "12-13 Stable; 14 Radioactive"
C-14 → carbon dating of ancient materials.

3. Allotropes — Diamond / Graphite / C-60

Allotrope	Structure / Conductivity	Key Facts / Uses
Diamond	3D tetrahedral; insulator	Hardest natural substance; transparent/lustrous; jewellery, glass cutters, sharp tools
Graphite	Hexagonal planar layers; conductor	Soft; layers slide; lubricant; pencil tip/electrode; source calls softest carbon form
Fullerene C-60	Football shape; 60 carbons	Buckminsterfullerene; semiconductor/lubricant/tumour research

TRICK - "Diamond INSULATES; Graphite CONDUCTS"
Opposite-looking exam trap.

EXAM TRAP - "C-60 = Kroto + Smalley + Curl"
Buckminsterfullerene discoverers — all three names.

4. Hydrocarbon Family — 2n+2 / 2n / 2n-2 / 2n-6

Type	Bond	Formula	Example
Alkane	Single C-C	C_nH_{2n+2}	CH4 Methane; C2H6 Ethane
Alkene	Double C=C	C_nH_{2n}	C2H4 Ethene/Ethylene
Alkyne	Triple C≡C	C_nH_{2n-2}	C2H2 Ethyne/Acetylene
Aromatic	Benzene ring	C_nH_{2n-6} — source formula	Benzene, Toluene, Naphthalene

TRICK - "ANE +2, ENE 0, YNE -2"
Alkane CnH2n+2; Alkene CnH2n; Alkyne CnH2n-2.

EXAM TRAP - "Acetylene = Ethyne = HC≡CH = C2H2"
Triple-bond identity trap.

5. Saturated vs Unsaturated

Type	Bond	Series / Example
Saturated	Only single C-C bonds	Alkane; Ethane H3C-CH3
Unsaturated	Double or triple bonds	Ethene H2C=CH2; Ethyne HC≡CH

TRICK - "SAT = SINGLE; UN-SAT = Double/Triple"
Alkane saturated; Alkene/Alkyne unsaturated.

6. Isomers vs Homologous Series

Concept	Meaning	Source Example
Isomers	Same molecular formula, different structure	C2H6O → CH3OCH3 and C2H5OH
Homologous Series	Same functional group; successive members differ by -CH2-	CH3OH → C2H5OH → C3H7OH

EXAM TRAP - "ISOMER = Same Formula, Different Structure"
Homologue = Same family + CH2 step.

7. Prefix Ladder — 1 to 10 Carbons

C atoms	Prefix	C atoms	Prefix
1	Meth-	6	Hex-
2	Eth-	7	Hept-
3	Prop-	8	Oct-
4	But-	9	Non-
5	Pent-	10	Dec-

TRICK - "Meth-Eth-Prop-But-Pent-Hex-Hept-Oct-Non-Dec"
1C se 10C tak prefix chain.

EXAM TRAP - "4C = But-, NOT Quad"
LPG main components Butane 4C + Propane 3C.

8. Suffix Rules — ane / ene / yne / anol

Class	Suffix	Example
Alkane	-ane	Methane, Ethane
Alkene	-ene	Ethene, Propene
Alkyne	-yne	Ethyne, Propyne
Alcohol	-anol	Methanol, Ethanol

TRICK - “Single ANE; Double ENE; Triple YNE; OH → ANOL”
Bond/functional-group suffix recall.

9. Functional Groups — OH / COOH / CHO / CO / NH2 / NO2 / X

Group	Name	Example
-OH	Hydroxyl / Alcohol	C_2H_5OH Ethanol
-COOH	Carboxyl / Carboxylic acid	CH_3COOH Acetic acid
-CHO	Aldehyde	HCHO Formaldehyde
-CO-	Ketone	CH_3COCH_3 Acetone
$-NH_2$	Amino	$C_2H_5NH_2$ Ethylamine
$-NO_2$	Nitro	$C_6H_5NO_2$ Nitrobenzene
-X (Cl, Br)	Halide	CH_3Cl Chloromethane

TRICK - “OH Alcohol; COOH Acid; CHO Aldehyde; CO Ketone”
Four most-tested functional groups.

EXAM TRAP - “Acetone = Ketone; Formaldehyde = Aldehyde”
Acetone CH_3COCH_3; Formaldehyde HCHO.

10. Methanol vs Ethanol vs Acetic Acid

Compound	Formula	Source Facts
Methanol / Wood Spirit	CH_3OH	Colourless; poisonous; excess consumption can cause blindness/death
Ethanol	C_2H_5OH	Liquid; good solvent; made by fermentation; used in tincture iodine/cough syrup
Ethanoic / Acetic acid	CH_3COOH	3-4% = vinegar; pure form = glacial acetic acid; MP 290 K

TRICK - “Methanol = Wood Spirit + Poison; Ethanol = Fermentation”
Vinegar = only 3-4% ethanoic acid.

EXAM TRAP - “Glacial Acetic = 99-100% Pure; MP 290 K”
Vinegar ≠ pure acetic acid.

11. Soap vs Detergent

Property	Soap	Detergent
Active group	$-COO^-$ carboxylate	$-SO_3^-$ sulphonate
Nature	Alkaline / Basic	Neutral
Hard water	No foam	Foams
Biodegradable	Yes	No — source note
Made from	Fats/oils by saponification	Petroleum derivatives

TRICK - “Soap = Saponification + Soft Water; Detergent = Does Hard Water”
Soap no foam in hard water; detergent foams.

EXAM TRAP - “Soap Alkaline; Detergent Neutral”
Source contrast.

12. Fuel Types — Solid / Liquid / Gas

Fuel Class	Source Examples
Solid	Wood, Coke, Coal, Charcoal
Liquid	Petrol, Diesel, Kerosene, Alcohol
Gaseous	LPG, CNG, Natural Gas

TRICK - “Solid WCCC; Liquid PDKA; Gas LPG-CNG-NG”
Fuel-type grouping.

13. LPG vs CNG — Butane/Propane vs Methane

Fuel	Full Form	Main Composition
LPG	Liquefied Petroleum Gas	Mainly Butane (C4) + Propane (C3)
CNG	Compressed Natural Gas	Mainly Methane (C1)

TRICK - “LPG = B+P; CNG = M”
LPG Butane+Propane; CNG Methane.

EXAM TRAP - “LPG ≠ Methane; CNG ≠ Butane”
Classic fuel composition trap.

14. Petrol / Diesel / Kerosene — Carbon Range

Fuel	Source Carbon Range
Petrol	C5-C10
Diesel	C10-C15
Kerosene	C10-C16

TRICK - “Petrol 5-10; Diesel 10-15; Kerosene 10-16”
Carbon-range number chain.

15. Octane vs Cetane

Number	Measures	Source Meaning
Octane Number	Petrol quality	Higher = better; less knocking
Cetane Number	Diesel quality	Higher = better combustion

EXAM TRAP - “O = Octane = Petrol; C = Cetane = Diesel”
Source also links Octane to 8C and Cetane to 16C in nomenclature notes.

16. Rapid-Fire Recall

Cue	Answer
Carbon atomic no./valency	6 / +4
Radioactive isotope	C-14
Hardest / conductor allotrope	Diamond / Graphite
Fullerene	C-60 Buckminsterfullerene
Alkane / Alkene / Alkyne	CnH2n+2 / CnH2n / CnH2n-2
Acetylene	Ethyne = C2H2
4-carbon prefix	But-
-CHO / -CO-	Aldehyde / Ketone
Wood spirit	Methanol CH3OH
Vinegar	3-4% ethanoic acid
Soap process	Saponification
LPG / CNG	Butane+Propane / Methane
Petrol / Diesel number	Octane / Cetane

17. Final Exam Traps — 12 Red Flags

- Carbon atomic no. 6, valency +4, Group 14 Period 2
- C-14 radioactive; C-12 and C-13 stable
- Diamond = electrical insulator; Graphite = conductor
- Buckminsterfullerene = C-60; Kroto-Smalley-Curl
- Alkane C_nH_{2n+2}; Alkene C_nH_{2n}; Alkyne C_nH_{2n-2}
- Acetylene = Ethyne = $HC\equiv CH$ = C_2H_2
- Isomer same formula/different structure; homologues differ by CH_2
- 4C prefix = But-, not Quad
- Acetone = ketone; Formaldehyde = aldehyde
- Methanol = poisonous wood spirit; Ethanol = made by fermentation
- Soap alkaline/no foam hard water; detergent neutral/foams hard water
- LPG = Butane+Propane; CNG = Methane; Octane petrol/Cetane diesel

EXAM TRAP - "6/4-C14-D/G-C60-Hydrocarbons-Acetylene-Iso/Homo-But-Acetone/Formal-Meth/Eth-Soap/Det-LPG/CNG/Oct-Cet"

Chapter ke 12 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Carbon = Z6, mass12.011, valency+4, Group14 Period2, covalent; catenation gives chains/branches/rings; C12/13 stable, C14 radioactive dating → Allotropes: Diamond 3D hard insulator, Graphite layered soft conductor/lubricant, Fullerene C60 football Buckminsterfullerene by Kroto-Smalley-Curl → Hydrocarbons: Alkane single C_nH_{2n+2} saturated, Alkene double C_nH_{2n}, Alkyne triple C_nH_{2n-2}, source aromatic C_nH_{2n-6}; Acetylene=Ethyne C_2H_2; isomer same formula diff structure, homologous series $+CH_2$ → Prefixes Meth-Eth-Prop-But-Pent-Hex-Hept-Oct-Non-Dec; suffix single-ane double-ene triple-yne OH-anol → Groups OH alcohol, COOH acid, CHO aldehyde, CO ketone, NH_2 amino, NO_2 nitro, X halide; Acetone ketone, HCHO formaldehyde aldehyde → Methanol CH_3OH wood spirit poison; Ethanol C_2H_5OH fermentation; Vinegar 3-4% CH_3COOH, Glacial 99-100% MP290K → Soap carboxylate alkaline saponification no foam hard water biodegradable; Detergent sulphonate neutral foams hard water source-not-biodegradable → Fuels: LPG=B+P, CNG=Methane; Petrol C5-10, Diesel C10-15, Kerosene C10-16; Octane=petrol quality, Cetane=diesel quality."

TOP 5: Diamond insulator vs Graphite conductor | Alkane/Alkene/Alkyne formulas | Acetone ketone vs Formaldehyde aldehyde | Soap vs Detergent | LPG/CNG + Octane/Cetane.

CHEMISTRY IN EVERYDAY LIFE — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 9 · Drugs + Dyes + Fibres + Glass + Cement/Fertilisers + Polymers + Rubber + Food Chemistry

Ye full notes nahi hai — Chapter 9 ke source mindmap se drugs, dyes, fibres, glass, cement/fertilisers, polymers, rubber/explosives aur food-chemistry ke names, uses, formulas, years aur exam traps ko fast recall format mein compress kiya gaya hai.

1. Drugs — Purpose + Classification

- Drugs = diagnosis, prevention and treatment of diseases
- Antacid → stomach HCl acidity control
- Antipyretic → body temperature/fever control
- Antihistamine → allergies, asthma, swelling, itching
- Tranquilizer → stress, anxiety, mental illness / sleep-related use
- Analgesic → pain killer
- Antibiotic → bacterial infection control
- Anaesthetic → sensory organs temporarily closed during surgery

TRICK - "A-A-A-T-A-A-A = Acid-Fever-Allergy-Stress-Pain-Bacteria-Surgery"
Antacid, Antipyretic, Antihistamine, Tranquilizer, Analgesic, Antibiotic, Anaesthetic.

2. Drug Examples — High-Frequency Pairs

Drug Type	Source Examples / Hook
Antacid	$Mg(OH)_2$, Cimetidine
Antipyretic	Paracetamol, Aspirin, Brufen
Antihistamine	Brompheniramine (Dimetapp), Terfenadine
Tranquilizer	Veronal; Equanil, Valium; Cocaine, Iproniazid; LSD, Mescaline; Barbiturates
Narcotic analgesic	Morphine, Heroin
Non-narcotic pain killer	Paracetamol, Dichlorofinac
Anaesthetic	Diethyl ether, Chloroform

TRICK - "MgOH2 Acid; Para/Aspirin Fever; Morphine/Heroin Narcotic"
Three direct exam hooks.

3. Antibiotics + Quinine + Sulpha

Item	Source Recall
Penicillin	First antibiotic; narrow spectrum; obtained from fungus
Chloramphenicol	Broad spectrum; Typhoid/high fever
Tetracycline	Broad spectrum
Quinine	Antimalarial; from Cinchona tree
Sulphanyl amide	First sulpha drug — 1908
Sulpha drugs	Sulphur + Nitrogen; prevent bacterial infections

TRICK - "Penicillin = Fungus + First + Narrow"
Chloramphenicol = Broad + Typhoid.

EXAM TRAP - "Quinine = Cinchona; Sulphanyl amide = 1908"
Natural-source + first-sulpha hook.

EXAM TRAP - "Aspirin = Fever + Pain; NOT Antibiotic"
Paracetamol = antipyretic + non-narcotic pain killer.

4. Antiseptic / Disinfectant

- Source framing: Antiseptics / Disinfectants prevent/destroy bacteria in open wounds
- Examples: Dettol, Iodoform (Iodine), Methylene blue

TRICK - “Dettol - Iodoform - Methylene Blue”
Open-wound bacterial control examples from source.

5. Dyes — Application + Chemistry

Type	Use / Example
Acidic dyes	Wool, silk, nylon; Methyl Orange, Congo Red
Basic dyes	Nylon, polyester; Yellow aniline, Green malachite
Bonding dyes	Alizarin — Red with Al, Blue with Ba
Azo dyes	Methyl Orange
Phthalein dyes	Phenolphthalein
Indigoid dyes	Indigo, Tyrian purple
Anthraquinone	Alizarin — natural
Triphenylmethane	Green malachite

TRICK - “Methyl Orange = AZO; Phenolphthalein = Phthalein; Indigo = Indigoid”
Alizarin = Anthraquinone + natural.

EXAM TRAP - “Alizarin: Al→Red, Ba→Blue”
Bonding-dye colour pair.

6. Fibres — Natural / Semi-Synthetic / Synthetic

Class	Examples / Source Use
Natural plant	Cotton, Flax/Linen
Natural animal	Wool — sheep; Silk — silkworm
Semi-synthetic — source list	Polyester, Metallic fibres, Rayon from cellulose
Nylon	Toothbrush bristles, ropes, parachutes, fishing nets, clothes
Polyester	Clothes, fire-fighting hosepipes
Carbon fibre	Spacecraft parts, sporting goods
Acrylonitrile	Synthetic blankets replacing wool

TRICK - “Nylon = Parachute + Toothbrush + Fishing Net”
Carbon fibre = Spacecraft; Acrylonitrile = Wool substitute.

EXAM TRAP - “Carbon Fibre ≠ Clothing”
Source specifically warns spacecraft parts, not clothing.

7. Glass — Base Facts

- Glass = amorphous homogeneous mixture of silicates of alkali metals
- Ordinary glass formula in source = $Na_2O \cdot CaO \cdot 6SiO_2$
- Ordinary glass = partially transparent

EXAM TRAP - “Glass = AMORPHOUS, not Crystalline”
High-frequency chapter trap.

8. Types of Glass — Best / Sunglass / Lab / Optics

Glass	Composition / Source Hook	Uses
Soda Glass	$Na_2CO_3 + CaCO_3 + SiO_2$	Test tubes, bottles, lab equipment, utensils
Photo Chromatic	Usual + Silver Chloride	Sunglasses
Jena Glass — Best	Zn + Barium Borosilicate	Chemical vessels, scientific instruments
Flint Glass	K carbonate + PbO + Silica	Bulbs, cameras, telescope/microscope lens, prism
Crookes Glass	Cerium oxide + Silica	Sunglass lenses
Pyrex Glass	Barium silicate + Na silicate	Lab equipment, pharmaceutical vessels
Quartz/Silica Glass	Melted SiO_2	UV lamps, reagent containers
Water Glass	Na_2CO_3 + Silica	Soluble in water; optical fibre, endoscopy — source note

TRICK - “JENA = BEST; Photochromatic = AgCl Sunglass; Crookes = Cerium Sunglass”
Pyrex = lab glass in source.

EXAM TRAP - “Water Glass = Soluble in Water”
Source note connects it with optical fibre/endoscopy.

9. Glass Colouring Agents

Agent	Colour
Carbon	Brown / Black
Cadmium Sulphide	Lemon Yellow
Cobalt Oxide	Dark Blue
Cuprous Oxide	Glitter Red
Cupric Oxide	Peacock Blue
Gold Chloride	Ruby Red
Manganese Dioxide	Pink
Ferric Oxide	Brown
K Dichromate	Green

TRICK - “Co = Dark Blue; AuCl = Ruby Red; $K_2Cr_2O_7$ = Green”
Most asked glass-colour pair hooks.

10. Cement + Fertilisers

Topic	Source Recall
Portland Cement	Joseph Aspdin — 1824
Cement CaO	60–70%
Cement SiO_2	20–25%
Al_2O_3 + Fe_2O_3	2–3%
N fertiliser	Urea, Ammonium sulphate
P fertiliser	Phosphatic asphalt — source term
K fertiliser	Potassium nitrate & sulphate
NPK	Nitrogen + Phosphorus + Potassium

TRICK - “Aspdin 1824 = Portland Cement; CaO 60–70%”
NPK = N + P + K.

11. Polymers — Monomer Pairs

Polymer	Monomer / Source
Polythene	Ethylene (CH_2=CH_2)
Polystyrene	Styrene
PVC	Vinyl Chloride
Bakelite	Formaldehyde + Phenol
Urea Formaldehyde Resin	Urea + Formaldehyde
Melamine	Melamine + Formaldehyde

TRICK - “Poly-Ethylene; PolySty-Styrene; PVC-Vinyl; Bakelite-Phenol+Formaldehyde”
Monomer-match chain.

12. Thermoplastic vs Thermosetting

Type	Heating Behaviour	Examples / Use
Thermoplastic	Softens; can remould/recycle	Polythene, PVC
Thermosetting	Does NOT soften; cannot remould	Bakelite, Melamine

TRICK - “ThermoPLASTIC = Pliable Again; ThermoSET = Set Forever”
Bakelite = heat/electric insulator; power switches.
Melamine = fire resistant; firefighter suits — source.

EXAM TRAP - "Bakelite = Thermosetting, NOT Thermoplastic"
One of chapter's biggest traps.

13. Rubber — Isoprene + Vulcanisation

- Natural rubber from vegetable milk (latex)
- Natural-rubber monomer = Isoprene
- Natural rubber: soft/sticky when hot; brittle when cold
- Synthetic rubber source examples: Neoprene = polymer of Chloroprene; Buna-N
- Vulcanisation improves rubber quality

Vulcanisation	Source Value
Temperature	313 K
Additive	Sulphur compound + ZnO
Tyre	5% sulphur
Battery case	30% sulphur
Result	Harder, stronger, more elastic, less sticky

TRICK - "Rubber = ISOprene; Vulcanise = 313 K + ZnO"
Tyre 5% S; Battery 30% S.

EXAM TRAP - "5 Tyre / 30 Battery"
Sulphur-percentage swap trap.

14. Explosives — Nobel vs Henning

Explosive	Source Detail
RDX	Cyclotrimethylene; Henning, Germany — 1899
TNT / TNG — source label	Trinitroglycerine (Nobel Oil); Dynamite by Alfred Nobel — 1863

EXAM TRAP - "Nobel 1863 = TNT/Dynamite; Henning 1899 = RDX"
Preserved exactly in source framing.

15. Food Chemistry — Preservatives + Antioxidants

Category	Source Examples
Food preservatives	Sodium benzoate, NaCl, Sugar, Oil, salts of sorbic & propanoic acid
Vitamin C	Ascorbic acid — fruits & vegetables
Vitamin E	Tocopherol — vegetable oil
Carotenoids	Orange/yellow fruits & vegetables
Polyphenolic antioxidants	Tea, Coffee, Chocolate, Soyabean
BHA / BHT	Synthetic antioxidants in packaged food

TRICK - "Preservative Star = Sodium Benzoate"
BHA/BHT = packaged-food synthetic antioxidants.

16. Artificial Sweeteners — 100 / 550 / 600 / 2000

Sweetener	Times Sweeter than Sucrose	Source Hook
Aspartame	100×	Widely used; successful
Saccharin	550×	Ortho-sulpho benzamide; not harmful
Sucralose	600×	Trichloro derivative of sucrose
Alitame	2000×	Strongest artificial sweetener

TRICK - "A-S-S-A = 100-550-600-2000"
Aspartame → Saccharin → Sucralose → Alitame.

EXAM TRAP - "Alitame = 2000× = Strongest"
Source says all listed sweeteners are for diabetics.

17. Final Exam Traps — 14 Red Flags

- Penicillin = fungus + first antibiotic + narrow spectrum
- Chloramphenicol = broad spectrum + typhoid
- Quinine = antimalarial from Cinchona tree
- Aspirin = antipyretic + pain killer, not antibiotic
- Methyl Orange = Azo; Alizarin = Anthraquinone; Phenolphthalein = Phthalein
- Nylon = parachute/toothbrush/fishing net; Carbon fibre = spacecraft
- Glass = amorphous, not crystalline
- Jena = best glass; Photochromatic = AgCl sunglass; Crookes = cerium sunglass
- Cobalt oxide = dark blue; Gold chloride = ruby red glass
- Bakelite = thermosetting; PVC = thermoplastic
- Natural rubber monomer = Isoprene
- Vulcanisation = 313 K + ZnO; tyre 5% S, battery 30% S
- Source: Alfred Nobel 1863 = TNT/Dynamite; Henning 1899 = RDX
- Sweeteners: Aspartame100, Saccharin550, Sucralose600, Alitame2000

EXAM TRAP - "Penicillin-Chlora-Quinine-Aspirin / Dyes / Fibres / Glass / Jena / Colours / Bakelite / Isoprene / 313K / Nobel-Henning / 100-550-600-2000"
Chapter ke 14 highest-yield traps isi chain mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Drugs: Antacid MgOH2, Antipyretic Para/Aspirin/Brufen, Antihistamine allergy, Tranquilizer stress, Narcotic Morphine/Heroin, Anaesthetic Ether/Chloroform; Penicillin fungus+first+narrow, Chloramphenicol broad+Typhoid, Quinine Cinchona, Sulphanyl amide1908 → Dyes: Methyl Orange Azo, Phenolphthalein Phthalein, Indigo Indigoid, Alizarin Anthraquinone natural + Al red/Ba blue → Fibres: Nylon parachute/toothbrush/net, Carbon fibre spacecraft, Acrylonitrile wool substitute → Glass amorphous; Jena best Zn+Barium borosilicate, Photochromatic AgCl sunglass, Crookes cerium sunglass, Pyrex lab; Co oxide dark blue, Au chloride ruby red → Cement Aspdin1824, CaO60–70%; NPK=N+P+K → Polymers: Polythene-Ethylene, PVC-Vinyl chloride, Bakelite-Phenol+Formaldehyde; Thermoplastic Poly/PVC vs Thermosetting Bakelite/Melamine → Rubber latex, monomer Isoprene; vulcanisation313K + ZnO, tyre5%S battery30%S → Explosives source: Nobel1863 TNT/Dynamite, Henning1899 RDX → Food: sodium benzoate preservative; antioxidants C/E/carotenoids/polyphenols/BHA-BHT; sweeteners A-S-S-A =100-550-600-2000, Alitame strongest."

TOP 5: Penicillin fungus/first/narrow | Jena best + Glass amorphous | Bakelite thermosetting | Isoprene + 313K + 5/30% S | Sweeteners 100-550-600-2000.

ENVIRONMENTAL CHEMISTRY — MEMORY TRICK SHEET

General Science · Chemistry · Chapter 10 · Pollution + Greenhouse + Smog + Acid Rain + Ozone + BOD + Bhopal + Green Chemistry

Ye full notes nahi hai — Chapter 10 ke source mindmap se environmental-pollution types, greenhouse gases, smog, acid rain, ozone depletion, water-pollution values, Bhopal tragedy, metal limits aur Green Chemistry ke exam facts ko fast recall tricks mein compress kiya gaya hai.

1. Environmental Pollution — 5 Types

- Environmental pollution = harmful substances se environment mein adverse changes
- Affects plants and animals
- Source pollutant examples: DDT, Plastics, Heavy Metals

Type	Core Source Hook
Air	Harmful gases/particulates; greenhouse gases, smog, acid rain, ozone depletion
Water	Industrial/agricultural/domestic waste; BOD, eutrophication, heavy metals
Soil	Pesticides, heavy metals, industrial waste; DDT persistent pollutant
Sound	Excessive noise; source says >85 dB harmful
Radioactive	Radioactive substances and nuclear waste

TRICK - "A-W-S-S-R = Air-Water-Soil-Sound-Radiation"
Five pollution types.

2. DDT + PCB — Persistent vs Carcinogenic

Pollutant	Full Form / Source Fact
DDT	Dichloro Diphenyl Trichloroethane — persistent pollutant
PCBs	Polychlorinated Biphenyls — carcinogenic industrial pollutant

EXAM TRAP - "DDT = STAYS; PCB = CANCER"
DDT persistent; PCBs carcinogenic.

3. Greenhouse Effect — Heat Trap

- Greenhouse gases trap heat in atmosphere → Earth temperature increases → Global Warming
- Source greenhouse list: CO_2, CH_4, N_2O, CFC (Freon), SO_2
- Source says main greenhouse gas = CO_2
- Source says most potent greenhouse gas = CH_4, 25× more potent than CO_2
- CFC = greenhouse gas + ozone depleter

TRICK - "CO_2 = MAIN; CH_4 = POTENT; CFC = DOUBLE ROLE"
CFC source role: greenhouse + ozone depletion.

EXAM TRAP - "Source Greenhouse List Includes SO_2"
Source greenhouse-gas list includes SO_2; preserve this source framing for revision.

4. Smog — General vs Photochemical

Smog	Formation / Composition	Nature
General	Smoke + Fog + SO_2; coal-burning industrial areas	Reducing
Photochemical	NO_x + Hydrocarbons + Sunlight; Ozone + NO + PAN	Oxidative

TRICK - "General = REDUCING; Photo = OXIDATIVE"
General: Smoke+Fog+SO_2.
Photo: NO_x+Hydrocarbon+Sunlight.

EXAM TRAP - “PAN = Peroxy Acetyl Nitrate”
Source formula: $CH_3CO \cdot O_2NO_2$; eye irritation + plant damage.

5. Acid Rain — 5.6 Rule

Fact	Source Recall
Normal rain pH	5.6
Acid rain	pH below 5.6
Responsible gases	SO_2 + NO_2
CO_2 role	Makes normal rain slightly acidic; source says not acid-rain cause
Damage	Marble/limestone monuments, soil, water bodies, forests
Marble cancer	Acid-rain erosion of Taj Mahal

TRICK - “Acid Rain = <5.6 + SO2/NO2”
NOT below 7.

EXAM TRAP - “CO2 → Normal Rain 5.6; SO2+NO2 → Acid Rain”
Source distinction.

6. Ozone Layer — Stratosphere + CFC

- Ozone layer present in Stratosphere
- Protects Earth from harmful UV radiation
- Main source cause of ozone depletion = CFC + Halogens
- CFC = Chlorofluorocarbon = Freon; compounds of C, Cl, F
- Used in refrigerators and AC
- Source says each CFC molecule can destroy thousands of ozone molecules
- Ozone hole first discovered over Antarctica

TRICK - “Ozone = STRATOSPHERE; CFC = FREON; Hole = ANTARCTICA”
CFC used in Refrigerator + AC.

EXAM TRAP - “CFC = Ozone Depleter + Greenhouse Gas”
Dual-effect trap.

7. BOD — Higher = Dirtier

- BOD = Biochemical Oxygen Demand
- Amount of O_2 consumed by aerobic bacteria to decompose organic matter in polluted water
- Higher BOD = more polluted

Water	Source BOD
Clean Water	5 ppm
Highly Polluted Water	17 ppm

EXAM TRAP - “BOD HIGH = Pollution HIGH”
Clean 5 ppm; Highly Polluted 17 ppm.

8. Eutrophication — Nutrient → Algae → O2 Down

- Excess nutrients = nitrates + phosphates from fertilisers/agricultural runoff/sewage
- Nutrients enter water → algal bloom
- Algae/decomposition consume dissolved O_2
- Dissolved O_2 depletes → fish/aquatic life die

TRICK - “N-P → ALGAE UP → O2 DOWN → FISH DOWN”
Nitrate + Phosphate enrichment chain.

EXAM TRAP - "Eutrophication = O2 DOWN"
Eutrophication = dissolved O2 DECREASE, not increase.

9. Drinking-Water Metal Limits — 0.005 to 5 ppm

Metal	Max Source Limit	Risk / Hook
Fe	0.2 ppm	Rust taste, staining
Mn	0.05 ppm	Neurological effects
Al	0.2 ppm	Possible Alzheimer link — source note
Cu	3.0 ppm	Liver/kidney damage
Zn	5.0 ppm	Nausea at high level
Cd	0.005 ppm	Kidney damage; Itai-Itai disease

TRICK - "Cd LOWEST 0.005; Zn HIGHEST 5.0"
Fe = Al = 0.2 ppm.

EXAM TRAP - "Cd = Itai-Itai"
Source disease association.

10. Bhopal Gas Tragedy — 3 Dec 1984

Fact	Source Recall
Date	3 December 1984
Location	Bhopal, Madhya Pradesh
Company	Union Carbide Limited
Gas	Methyl Isocyanate (MIC)
Formula	$H_3C-N=C=O$
MIC use	Pesticide manufacturing

TRICK - "3-12-84 + MIC + Union Carbide"
Bhopal master memory.

EXAM TRAP - "1984 Bhopal, NOT 1986"
Source contrasts 1986 with Chernobyl.

11. Green Chemistry — Prevention over Cure

- Green Chemistry = chemical products/processes designed to reduce or eliminate chemical hazards
- Aim = pollution prevention at source
- Source says uses water as solvent: cost-effective, non-flammable, non-carcinogenic
- 12 principles of Green Chemistry
- Old Zapper technology developed by TERI
- TERI = The Energy and Resource Institute

TRICK - "GREEN = Hazard DOWN at SOURCE"
Prevention over cure; source solvent = Water.

EXAM TRAP - "TERI → Old Zapper"
The Energy and Resource Institute.

12. Abbreviations Flash

Abbrev.	Full Form
DDT	Dichloro Diphenyl Trichloroethane
CFC	Chlorofluorocarbon (Freon)
PAN	Peroxy Acetyl Nitrate
BOD	Biochemical Oxygen Demand
PCBs	Polychlorinated Biphenyls
MIC	Methyl Isocyanate
TERI	The Energy and Resource Institute

TRICK - "D-C-P-B-P-M-T"
DDT-CFC-PAN-BOD-PCB-MIC-TERI abbreviation chain.

13. Number Flash — 5.6 / 5 / 17 / 0.005 / 5 / 1984

Number	Meaning
5.6	Normal-rain pH; acid rain below this
5 ppm	Clean-water BOD — source
17 ppm	Highly polluted-water BOD — source
0.005 ppm	Cd max limit — lowest
5.0 ppm	Zn max limit — highest
3 Dec 1984	Bhopal gas tragedy

TRICK - "5.6 - 5 - 17 - .005 - 5 - 3/12/84"
Chapter number-lock chain.

14. Final Exam Traps — 12 Red Flags

- Acid rain pH = below 5.6, not below 7
- Acid rain source cause = SO_2 + NO_2; CO_2 linked to normal rain pH 5.6
- General smog = reducing; Photochemical smog = oxidative
- PAN = Peroxy Acetyl Nitrate
- Ozone depletion = stratospheric pollution
- CFC = Freon; refrigerators/AC; ozone depleter + greenhouse gas
- Bhopal = 3 December 1984; MIC; Union Carbide Limited
- Higher BOD = more polluted
- Source BOD: clean 5 ppm; highly polluted 17 ppm
- Eutrophication = dissolved O_2 decreases
- Cd limit 0.005 ppm lowest; Zn 5.0 ppm highest; Fe=Al=0.2 ppm
- PCBs = carcinogenic; TERI = The Energy and Resource Institute

EXAM TRAP - "5.6-SO2NO2 / Smog / PAN / Strato / CFC / 3-12-84 / BOD↑ / 5-17 / O2↓ / Cd-Zn / PCB-TERI"
Chapter ke 12 highest-yield traps isi order mein revise karo.

MASTER MEMORY CHAIN — Poora Chapter Ek Saath

MASTER - 1-Line Revision Chain

"Pollution 5 types A-W-S-S-R; DDT persistent, PCBs carcinogenic → Greenhouse effect traps heat; source gases CO_2-CH_4-N_2O-CFC-SO_2, main CO_2, potent CH_4, CFC dual role → Smog: General=Smoke+Fog+SO_2 reducing; Photo=NO_x+Hydrocarbon+sunlight oxidative with Ozone+NO+PAN; PAN=Peroxy Acetyl Nitrate → Acid rain <5.6, gases SO_2+NO_2; normal rain 5.6 due dissolved CO_2; marble cancer/Taj damage → Ozone stratosphere, CFC=Freon=C/Cl/F, fridge+AC, hole Antarctica → BOD = Biochemical Oxygen Demand; HIGH BOD = MORE pollution; source clean5 ppm vs highly polluted17 ppm → Eutrophication Nitrate+Phosphate → Algal bloom → dissolved O_2 ↓ → fish death → Metal limits: Cd0.005 lowest, Zn5 highest, Fe=Al0.2; Cd-Itai-Itai → Bhopal 3 Dec1984, Union Carbide, MIC H_3C-N=C=O, pesticide manufacturing → Green Chemistry = hazards reduce/eliminate at source, source water solvent, 12 principles; TERI = The Energy and Resource Institute → Old Zapper."

TOP 5: Acid Rain <5.6 + SO_2/NO_2 | General reducing vs Photo oxidative | Ozone Stratosphere + CFC/Freon | BOD high = more pollution | Bhopal 3 Dec 1984 = MIC + Union Carbide.